TEAM COMPILED RESEARCH PAPER

Optimal Sensor Placement for Wastewater Network Monitoring

Research findings, data availability, methods, and evaluation framework for the SA Waterproject

Project: Optimal Sensor Placement for Wastewater Network Monitoring, SA Water
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Prepared: 16 October 2026

This paper brings the team's literature, sensor technology review, placement methods, sewer-condition evidence, South Australian data assessment, and current project decisions into one research record. It is organised around the wastewater monitoring problem and the decisions required for a defensible placement method.

Abstract

The project addresses the following operational question: given a wastewater sewerage network, where should a small number of sensors be placed to obtain high monitoring capability while controlling deployment, telemetry, maintenance, and operating cost? The team has combined a question-led review of sensor technologies, a comparative study of optimisation methods, a review of sewer condition and failure factors, and a survey of openly accessible wastewater network data in South Australia. The evidence shows that sensor placement is not a single "coverage" problem. The useful location depends on the required response: continuous level trends, threshold overflow alarms, flow reconstruction, contaminant detection, hydrogen sulfide control, condition assessment, or remote operational awareness. It also depends on network direction, hydraulic behaviour, environmental sensitivity, failure risk, access, power, telemetry, and maintenance.

The reviewed methods range from engineering heuristics, complex-network centrality, mutual information, and value of information to evolutionary greedy optimization, elevation-conditioned coverage models, topology-derived flow inference (TDFI), spatiotemporal graph neural network (GNN) reconstruction, unsupervised deep graph clustering, GIS-based fuzzy Analytical Hierarchy Process (AHP) candidate screening, and joint Bayesian calibration via optimal experimental design. Published case studies show that well-placed sensors consistently outperform random or simple heuristic layouts, but reported percentages are network-specific and cannot be transferred directly to SA Water. Furthermore, operational evidence highlights acute trade-offs between battery-preserving telemetry cadences and high-frequency anomaly detection, as well as the need to balance warning lead time against utility triage capacity. South Australian public data are sufficient for an initial graph and attribute benchmark in one metro sewer extract, but they do not provide a complete catchment, operational incidents, historical sensor readings, maintenance-hole depth, or dependable roughness and condition data. The research therefore supports a staged, purpose-specific method: integrate client network and environmental data; build the directed graph and candidate set; define monitoring objectives and priority weights; score hydraulic, information, risk, and operational value; select sensor type and location jointly under physical and budgetary constraints; and compare baselines while stress-testing deployments against dropouts, missing data, and telemetry latency. The final sensor mix, density, and coverage remain outputs to be evaluated on the supplied SA Waternetwork rather than assumptions imported from other jurisdictions.

Keywords: wastewater network; sewer monitoring; optimal sensor placement; level sensing; flow reconstruction; graph neural networks; evolutionary optimization; information theory; risk-based monitoring; South Australia; SA Water.

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1 Project purpose, scope, and boundaries

1.1 The operational problem

Wastewater collection networks are spatially distributed, mostly underground, and exposed to changing hydraulic, environmental, and operational conditions. A failure can present as surcharge, reduced capacity, overflow, pollution, odour, corrosion, flooding, or a service interruption. A sensor strategy is useful only when it turns a measurement into an operationally relevant action. The project is therefore concerned with the location and density of monitoring assets, but also with the question each asset is expected to answer.

The project brief defines the central problem as finding the optimal locations for float and level sensors so that a minimum number of devices provides high monitoring capability across a sewerage network. The intended value is earlier detection and prevention of overflows or overcapacity, improved monitoring of sewer channels, protection of environmentally sensitive receiving areas such as rivers and ponds, and lower avoidable maintenance cost [PG S2-22 project team, 2026c,b]. Existing sensors are retained as constraints or baseline assets; the research does not assume that every existing installation should be replaced.

The current focus is sewer level monitoring and the wider placement framework needed to support it. Other sensors remain relevant where they answer a distinct operational question, but the work does not treat a level sensor as a universal substitute for flow, water-quality, gas, or condition monitoring.

1.2 Scope boundaries

Table 1: Project boundaries carried into the compiled research paper.

Area	Included	Boundary or interpretation
Network type	Wastewater collection and sewerage networks, including gravity networks and relevant pump or rising-main interfaces.	Drinking-water supply and stormwater are not the primary target. Stormwater methods are included only as transferable methodological evidence.
Monitoring purpose	Level trends, surcharge and overflow warning, capacity awareness, environmentally sensitive locations, and selected flow, contaminant, gas, and condition questions.	Each purpose requires its own sensor and validation logic.
Placement decisions	Candidate locations, sensor density, sensor type, existing deployments, cost, access, telemetry, power, maintenance, and staged rollout.	A numerical recommendation is not final until client network and operational data are available.
Data	Network geometry, inverts, surface elevation, diameter, age, material, grade, incidents, historical failures, sensor readings, hydraulic models, and environmental context.	Public substitutes can support benchmarking but do not establish the full operational truth.
Physical work	Research, data integration, modelling, prototype algorithm, evaluation, and handover documentation.	Physical entry, installation, commissioning, and operational approval are outside this research stage [PG S2-22 project team, 2026b].

1.3 What this paper records

This is one team research record rather than a set of separate member reports. The findings are grouped by the decisions they support:

1. which sensor technologies can answer each wastewater monitoring question;
2. which placement algorithms can use topology, hydraulics, information, risk, or failure evidence;
3. which sewer condition factors should influence risk and monitoring priority;
4. what South Australian public data can and cannot support;
5. how the evidence can be combined into a SA Waterevaluation method; and
6. what remains to be obtained, tested, or approved before a client-facing recommendation.

The paper distinguishes published study results from the proposed SA Waterworkflow. Results from Hong Kong, Europe, the United States, and other catchments demonstrate methods; they are not presented as measurements of the SA Waternetwork.

2 Evidence base and project research method

2.1 Evidence brought together

The compiled evidence consists of the team's sensor-technology paper, placement and optimisation reading notes, sewer-condition and failure studies, data availability reports, project charter and scoping material, the stakeholder presentation, and the underlying papers stored in the project research library. The literature covers sensor hardware, sewer hydraulics, contamination and condition, information theory, complex networks, evolutionary optimisation, machine learning, matrix completion, sparse sensing, and risk-based set cover.

The data work follows a separate path from the literature work. Public ArcGIS services were queried anonymously where possible; visible service definitions and access restrictions were recorded; network fields were checked for usable values rather than field names alone; and graph properties were calculated from pipe inverts and endpoints. No credentials were used or sought during the public survey. The access-restricted endpoints were documented as data-request targets, not treated as if they had been downloaded [Paul Jaison, 2026].

2.2 Research activities

Table 2: Research activities and the decisions they support.

Activity	Method	Output for the project
Sensor review	Organise technologies by operational question, measurement principle, data product, installation conditions, maintenance, power, telemetry, and limitations.	A sensor-selection basis for level, flow, water quality, gas, condition, and data transmission.
Placement-method review	Compare engineering, information-theoretic, clustering, topology, observability, optimisation, ML, and sparse-sensing methods.	A method map showing required data, objective, algorithm, metric, and transfer risk.
Condition-factor review	Identify physical, environmental, climate, and operational factors associated with defects and deterioration.	A feature catalogue for risk scores, prioritisation, and sensitivity tests.
Public data survey	Query open service endpoints; inspect layers, fields, counts, ranges, completeness, geometry, and network direction.	A benchmark data inventory and a precise request plan for restricted SA Waterdata.
Cross-source synthesis	Align sensor purpose, network graph, failure risk, data availability, and evaluation measures.	A proposed staged workflow that can be run when client data and objectives are confirmed.
Project planning	Use two-week sprints, Jira, Teams, regular academic supervision, and industry meetings to manage literature, data, method, prototype, evaluation, and handover deliverables.	A staged route from research evidence to a reproducible prototype and client review [PG S2-22 project team, 2026a,b].

2.3 Reading the quantitative findings

Reported values such as mutual-information gain, reliability, classification accuracy, reconstruction error, risk coverage, and NSE are retained because they explain how each method was tested. They must be read with the network size, input data, sensor count, objective, and model assumptions beside them. A method that obtains 80% coverage in a directed network may be measuring upstream connectivity; a method that obtains 0.95 NSE may be reconstructing a simulated flow field; and a method that predicts 85% of condition cases may rely on CCTV-derived labels from a different sewer system. None of these quantities is a SA Watertarget until it is reproduced on the client network.

3 Wastewater network representation and decision variables

3.1 Network graph

A collection network can be represented as a directed graph

$$G = (V, E),$$

where V contains maintenance holes, junctions, pump stations, outfalls, shafts, or other monitoring-relevant nodes, and E contains gravity mains, rising mains, or other directed links. In a gravity network, the pipe

endpoint geometry and the field convention for flow direction must be interpreted together before the graph is oriented. A sensor placed at node i may observe or constrain states at upstream nodes whose flow passes through i; the relevant set is not the Euclidean neighbourhood alone.

For a sensor set $S \subseteq V$, a simple upstream coverage term can be written as

$$C(S) = \sum_{j \in V} w_j \, \mathbf{1}(\exists i \in S : j \in U(i)) \,,$$

where $U(i)$ is the upstream set of candidate location i, w_j is a priority weight, and the indicator avoids counting a node twice. The weight may represent population, environmental sensitivity, failure risk, overflow consequence, or a documented monitoring priority. This expression is a proposed project representation, not an observed SA Waterresult.

3.2 Attributes needed for a useful model

The literature and data survey repeatedly distinguish between fields that describe a network and fields that make a placement decision defensible:

Table 3: Network information and how it enters placement.

Information	Role	Common problem
Pipe endpoints and direction	Builds the directed graph and upstream/downstream reachability.	Field labels may describe the upstream and downstream ends by flow rather than by digitised geometry; clipped administrative boundaries may split an analytical catchment.
Invert and surface elevation	Supports grade, depth, surcharge interpretation, and hydraulic reconstruction.	Public data often contain one elevation but not the other; contour interpolation can reconstruct depth, but the interval and validation error must be reported.
Diameter, shape, length, material, age, slope	Describes hydraulic capacity, deterioration, cost, and condition risk.	Material and internal diameter may be unknown; nominal diameter is not always hydraulic internal diameter.
Hydraulic time series or model	Supports level, flow, travel time, contaminant spread, and observability calculations.	Historical readings and calibrated models are often unavailable; simplified models can make placement look more certain than it is.
Incident and failure history	Locates recurrent choke, overflow, service, root, infiltration, or structural risk.	Service-request and maintenance layers may be restricted or incomplete.
Receiving environment and land use	Weights consequences near rivers, ponds, sensitive ecosystems, residences, roads, and industrial sites.	Environmental and social layers require spatial matching and clear governance.
Asset and deployment constraints	Represents access, safety, power, telemetry, installation geometry, existing devices, maintenance, and cost.	Purchase cost alone does not represent total life-cycle cost.

3.3 Monitoring objectives are not interchangeable

The same sewer node can be valuable for different reasons. A downstream node may provide broad upstream reach and early warning of accumulated flow, while an upstream node may preserve a concentrated contaminant signal. A main-path or outfall location may be highly informative for network-wide state reconstruction but not identify a local blockage. A high-risk pipe may merit monitoring even if it adds little information to a flow reconstruction model. The placement objective must therefore be stated before the algorithm is chosen.

The project evidence supports at least six objective families:

1. **hydraulic state**: estimate level, flow, velocity, surcharge, or capacity at unmonitored locations;
2. **event detection**: detect overflow, rapid rise, blockage, or abnormal downstream behaviour;
3. **source or contaminant detection**: identify where a contaminant entered and whether it reached a sensor;
4. **condition and failure risk**: prioritise assets with greater probability or consequence of deterioration;
5. **environmental protection**: increase monitoring near sensitive receiving environments or high-consequence locations; and
6. **operational deployment**: maximise useful information after access, power, telemetry, maintenance, and existing-asset constraints.

4 Sensor technologies and what each can measure

4.1 Technology selection principle

The sensor review reaches a simple conclusion: different technologies answer different questions. Level is not flow; a threshold event is not a continuous trend; a gas headspace measurement is not dissolved hydrogen sulfide; and a condition inspection is not a live hydraulic measurement. The useful technology is the one that produces enough reliable information for the intended response at a maintainable site.

Table 4: Question-led sensor technology map.

Question	Technologies	Useful output	Important limitations
How full is the sewer?	Radar; ultrasonic; LiDAR or time-of-flight; float or tilt.	Continuous level trend, rate of rise, surcharge or threshold event.	Condensation, humidity, mist, foam, solids, obstructions, mounting geometry, fouling, and line-of-sight affect non-contact devices; float or tilt does not provide a continuous trend.
How much is flowing?	Contactless ultrasonic flow; level plus geometry; pump drawdown; hydraulic model.	Flow rate, velocity, loading, capacity, or pump performance.	Partially full flow requires wetted area and velocity; direction, backwater, geometry, multiple inlets, calibration, and site conditions matter.
What is in the wastewater?	pH, electrical conductivity, temperature, turbidity, UV/Vis, fluorescence, nutrients, dissolved oxygen, microbial fuel cell.	Chemical change, illicit-discharge clue, organic-load proxy, toxicity, or process indicator.	Fouling, matrix effects, interference, drift, turbidity, and local calibration limit generalisation; several optical variables are proxies rather than universal direct concentrations.
Is hydrogen sulfide present?	Headspace gas sensors; dissolved liquid H_2S sensors; UV/Vis or electrochemical devices.	Odour and corrosion risk, septicity, dosing or treatment control.	Gas and liquid phases need different locations, ranges, calibration, maintenance, and interpretation.
Is the asset blocked or damaged?	CCTV, acoustic, computer vision, fibre optic.	Structural defect, root, deposit, joint, deformation, or blockage evidence.	Inspection can be detailed but is not the same as continuous hydraulic monitoring; access, flow, images, training data, and skilled review are constraints.
Can readings be used remotely?	LoRaWAN, cellular, gateways, data platforms, soft sensors.	Remote telemetry, alerts, stored series, or model-based estimates.	Radio coverage, battery, sampling interval, data quality, cybersecurity, and model validation affect operational value.

4.2 Level sensing

4.2.1 Radar and continuous level

Radar is a non-contact level technology that can support continuous trends, rate-of-rise analysis, surcharge detection, and abnormal pattern recognition. The evidence reviewed describes radar as less affected by humidity and mist than ultrasonic sensing, but not immune to poor mounting, condensation, surface conditions, or fouling. A Drenoyanis et al. prototype for Shoalhaven Water combined a level device with LoRaWAN communication; the reported communication range was up to about 20 km in an urban setting and 35 km line-of-sight. These are radio-link observations, not radar sensing ranges, and the prototype did not establish a general field accuracy or a validated choke-prediction rule [Drenoyanis et al., 2019].

The practical selection questions are the level range, required accuracy and time resolution, installation geometry, power budget, radio path, access, fouling exposure, and maintenance interval. A radar measurement can be valuable for a trend-based model only if its mounting and data-quality checks are designed with the same care as the placement algorithm.

4.2.2 Ultrasonic, LiDAR, and time-of-flight

Ultrasonic sensors are attractive because they are non-contact and can be low power, but steam, mist, condensation, foam, temperature-dependent sound speed, and obstructions can distort the echo. Do et al. used 62 one-year wastewater level datasets from Stonyfell, South Australia, and showed that changes from a learned normal level pattern could support early identification of a choke-like abnormality [Do et al., 2023]. The finding is useful for anomaly detection, but it does not establish a universal threshold or a placement density.

Utepov et al. compared Sharp IR, TF-Luna LiDAR, TOF400, and JSN-SR04T in an imitative sewer-well prototype. The reported JSN-SR04T performance was approximately ±5 cm up to 2 m and TF-Luna approximately ±8 cm up to 5 m in the tested setup. The complete prototype material cost was about USD 170.07 excluding labour. LiDAR and other optical devices need a clear path and can be affected by condensation, fouling, and wastewater surface conditions. These results support low-cost prototyping, not an assumed field-node price or service life [Utepov et al., 2023].

4.2.3 Early detection of sewer chokes and overflows

In sewer level monitoring, early warning requires separating genuine developing blockages from normal diurnal variations and wet-weather peaks without creating an unmanageable false-alarm burden. The SA Watertrial in Stonyfell, Adelaide, highlights both the potential and the operational constraints of continuous acoustic level sensing [Do et al., 2023]. The trial initially deployed 82 non-contact ultrasonic level sensors, yielding 62 operational datasets over approximately one year across 11.3 km of gravity network in a suburb selected for higher-than-average choke incidence. Initial fixed thresholds of 150 mm and 300 mm generated severe operational alarm fatigue: the 150 mm threshold alone produced over 100,000 low-priority alarms in 2019, swamping operational response teams.

To resolve this, Do et al. [2023] developed the Finding Irregular Duration (FID) algorithm, which detects chokes by identifying irregular level peaks accompanied by persistent, unusually elevated durations rather than instantaneous depth spikes. FID compares each measured level against a local 14-day rolling historical baseline within a two-hour moving window. In the Stonyfell deployment, requiring more than 20 minutes of sustained abnormal duration at depths exceeding 50 mm enabled FID to successfully issue proactive warnings for all 24 confirmed blockage events, achieving lead times ranging from 8 to 57 days (mean lead time of 32.6 days) prior to any surface spill. However, FID is an anomaly-detection algorithm that presupposes an established monitoring point; it is not a sensor-placement optimization method. Furthermore, its empirical parameters reflect specific pipe diameters and baseflows in Stonyfell and do not establish a universal threshold or placement density.

At a broader utility scale, this early-warning paradigm has been validated by Sydney Water's active network sensing programme under the Wastewater Overflow Model (WWOM) synthesis report [Sydney Water, 2024]. Sydney Water deployed over 8,000 IoT level monitors (scaling toward 15,000) across metropolitan reticulation manholes, identifying 300 to 370 developing blockages per month before sewage spills could occur. This active choke interception avoids an estimated $400,000 per month in reactive containment and environmental remediation costs [Sydney Water, 2024, Drenoyanis and Prackwieser, 2022]. These findings confirm that early choke warning demands continuous, high-fidelity level trends, but place a strict operational requirement on sensor placement: candidate locations must balance hydraulic sensitivity with low false-alarm generation and field accessibility for preventative clearing.

4.2.4 Float and tilt sensors

Float or tilt devices can provide a robust threshold event, such as a high-level alarm, with a simple sensing principle. They do not provide the level history, rate of rise, or normal-pattern information available from a continuous sensor. Grease, rags, solids, and mechanical obstruction can prevent movement or create a false state, so a float strategy needs inspection and cleaning arrangements [United States Environmental Protection Agency, 1999]. The stakeholder brief also indicates that continuous level sensors may cost several times more than float or pulse devices; this is a project-specific stakeholder statement and must be replaced by verified procurement and life-cycle figures before a cost model is finalised [PG S2-22 project team, 2026b].

4.2.5 Blockage detection and sensor spacing from level

A level series carries more than fill state. Li et al. showed that a blockage produces a two-sided signature: sewage is intercepted, the level rises upstream of the obstruction and falls downstream, and the direction of the change tells an instrumented manhole which side of it the obstruction lies on. The study built a double-threshold discriminant from that signature. If the rate of level change exceeds a positive threshold and the range of change exceeds a second positive threshold at the same time, a blockage downstream of the gauge is declared; if the rate falls below a negative threshold and the range below a second negative threshold, a blockage upstream is declared; otherwise the reach is treated as normal. The thresholds were derived empirically from experiment rather than from a hydraulic model.

Two scales were used. The pilot rig was PVC pipe at DN150 to DN300 with pumped flow regulated by an outlet valve, level sensors of 1 mm accuracy upstream and downstream of the blocking point, 15 s sampling and 13 min per experimental group; flow was varied from 0.5 to 5 m^3/h, chosen from a daily consumption of 40 L per inhabitant to represent a population range of 130 to 1,300. The full-scale experiment installed six level sensors at manhole wellheads on a rural network. Blocking objects of 0.1, 0.15 and 0.2 kg dry weight were introduced at known manholes under both low and peak consumption.

Abnormal level rise was detectable within one minute and within 4 m upstream of the blocking point, reaching a peak 300% to 700% above the pre-blockage level within four minutes, with the rise positively related to the degree of blocking. The identification rate was 95.3% within 4 m upstream and 93.3% downstream; a 0.2 kg blockage was identified in every trial at 29.55 m and 60.4 m downstream, while 0.15 kg and 0.1 kg gave 93.3%. Overall success rates including normal operation were 80.4% for the DN200 pilot and 95.6% for the DN300 full-scale case.

The result the project should carry forward is a density constraint rather than a detection method. A sensor detected a level change caused by a blockage 180 m upstream of itself, but the authors recommend that spacing between monitoring points should not exceed 100 m if identification performance is to hold. That is an empirically grounded upper bound on sensor spacing for small-diameter gravity sewers and it sets a ceiling on how sparse an optimised layout can honestly be. The transfer limits are substantial: the thresholds are specific to these diameters and flow ranges with no stated recalibration procedure, the pilot rig used pumped clean water so sediment, grease and solids behaviour are absent, blockages were introduced by dry weight rather than allowed to develop, and 20% of DN200 pilot blockages and 16.7% of full-scale post-blockage level drops were missed [Li et al., 2023].

4.3 Flow measurement and flow reconstruction

For a partially full sewer, flow depends on wetted area and mean velocity. A level reading alone is not a flow measurement unless pipe geometry, direction, hydraulic assumptions, and calibration support the conversion. The level-to-flow relation can change under backwater, sediment, multiple inlets or outlets, variable roughness, and non-uniform flow.

Alshami et al. tested contactless ultrasonic flow and level measurements at 12 Hong Kong locations. The level-flow correlation was 0.36 and the average relative flow error was 72.14% against Manning-equation estimates. The study associated the error with small pipes, installation angle, flow direction, level, and multiple inlets and outlets. The result is a direct warning against treating a contactless sensor as accurate by technology name alone; site-specific calibration and hydraulic interpretation are required [Alshami et al., 2023].

Level-derived Manning estimates are useful for a benchmark where geometry and roughness are known, but roughness is often missing or assumed. Pump drawdown can estimate pumped volume or rate but describes a different hydraulic situation from gravity conveyance. A project flow model should therefore report the measurement path: direct velocity, level-plus-geometry, pump drawdown, or model reconstruction.

A soft sensor offers a third measurement path alongside direct velocity, level-plus-geometry and model reconstruction. Lin et al. estimated flow rate from a time series of water levels measured in a single manhole using a multilayer perceptron with one hidden layer of 128 neurons. The design responds to a specific difficulty: flows into and out of a manhole may be free-surface or pressurised and subcritical, critical or supercritical, which makes the depth-to-flow relation non-bijective and even multivalued, so one depth can correspond to two or more flow rates and no rating curve can resolve it. The authors' answer is to supply depths from both before and after the target instant, on the reasoning that the flow rate at a given time is influenced by earlier depths through the domain of dependence of the Navier–Stokes equations and influences later depths through the domain of influence. Training used three manholes with different hydraulic characteristics in urbanised areas of China at 1-minute resolution across 2021, plus one manhole simulated by a three-dimensional CFD model. The optimal input window was about four minutes, a 1-minute sampling frequency was sufficient, and windows of 12 minutes or more degraded performance. On the simulated manhole the baseline mean relative error was 2.88% with NSE 0.9372, rising only to 3.26% when the assumed 95% level-sensor confidence interval was widened from 3 mm to 10 mm, and to 3.38% with noise added to both level and flow. Bias behaved differently from noise: a −30 mm level bias present in training was absorbed, but the same bias applied only at test time turned a 9.00% input error into 11.80% output error, and a bias of 26.18% in the training flow data produced 37.62% error, an amplification of 34.55%. The quality of the flow data used for training is therefore critical, and the method requires a flow campaign at every manhole covering dry weather and a range of wet-weather conditions [Lin et al., 2025a].

A follow-up study attacked that training-data requirement by reading depth at two adjacent manholes instead of one, and compared three model structures on the same input: the Saint-Venant equations solved with an HLL Riemann solver and only two calibrated parameters, Manning roughness and sediment depth; the same multilayer perceptron; and a physics-informed neural network. Case Study 1 was a 600 mm pipe of slope 0.005 and length 70 m, 38 days and 52,026 samples covering dry and wet weather with average dry-weather inflow near 210 m^3/d; Case Study 2 was a 1,200 mm pipe of slope 0.005 and length 113 m, two weeks and 20,075

samples, dry weather only. On the real pipes the perceptron was the more accurate, with R^2 of 0.945 and 0.914 against 0.931 and 0.754 for the Saint-Venant sensor, the gap widening during wet weather and at low flow. The data-efficiency comparison is the result that matters for deployment cost: the Saint-Venant sensor reached 14.7% mean relative error trained on three hours of data and 13.0% on one day that contained no wet-weather flow at all, improving only to 11.8% at 35 days, whereas the perceptron needed 54.7% at three hours, 5.60% at 11 days and 5.09% at 35 days, with each rain event in the training window producing a step improvement. The Saint-Venant calibration also returns physically meaningful parameters, recovering roughness and sediment depth statistically identical to the true values in simulation and estimating 5.2 ± 0.9 mm and 31.9 ± 1.2 mm of sediment in the two real pipes, which makes the same device a sediment-accumulation indicator as well as a flow estimate. Both models ran on a Raspberry Pi 5 with a LoRa link between the two manholes, producing an estimate in 0.0007 s and 1.5 s respectively. The physics-informed network failed outright: it has no separate training phase, so training was always terminated by the 60 s sampling interval rather than by convergence, giving R^2 of −1.334 and −0.324 and mean relative errors of 97.58% and 163.18%. The transfer limits are that both real pipes are short and already carried flow meters at both ends, and that Case Study 2 contains no wet weather, so the data-efficiency claim rests on a single pipe [Lin et al., 2025b].

4.4 Water-quality and optical sensing

Low-cost level, temperature, and electrical-conductivity devices were operated for more than a year in a Melbourne catchment to examine dry-weather patterns and short events. The study did not use pH in that sensor set; a separate three-year raw-sewer programme used temperature, electrical conductivity, turbidity, and pH with cleaning and self-diagnostic provisions [Shi et al., 2022, Li et al., 2017]. These examples show how context is needed: a conductivity change can be a useful illicit-discharge clue, but it does not identify a source without spatial, temporal, and chemical interpretation.

UV/Vis spectroscopy can provide reagent-free estimates of suspended solids and total or dissolved chemical oxygen demand. Fluorescence features such as Peak T and Peak C may correlate with BOD, COD, or TOC. Optical signals are affected by overlapping fluorophores, inner-filter effects, turbidity, fouling, and wastewater matrix. They should be treated as locally calibrated proxies unless a validated calibration establishes a direct relation for the target stream [Lepot et al., 2016, Carstea et al., 2016].

Nutrient sensors, dissolved-oxygen sensors, and microbial-fuel-cell approaches may support targeted investigations. Selectivity, membrane behaviour, temperature, drift, interference, cleaning, and calibration are central design variables. Their value is highest where a known chemical or process question justifies the extra maintenance.

4.5 Hydrogen sulfide

The project separates two distinct H_2S questions:

1. **headspace gas**: odour, worker exposure, corrosion atmosphere, and peak gas concentration; and
2. **liquid dissolved H_2S**: septicity, sulfide loading, dosing, and treatment control.

Pacheco et al. compared commercial gas sensors at two sewer sites, including OdaLog/MyData with a 0–200 ppm range and SulfiLogger with a 0–1000 ppm range. The compared specifications showed materially different battery durations under their stated transmission conditions, from months to approximately two years depending on model and schedule. Agreement was better at lower concentrations and deviations increased at higher concentrations; humidity and management practice also affected readings [Pacheco Fernández et al., 2021]. A gas range, calibration regime, logging interval, and maintenance plan must be chosen for the actual site.

Despot et al. examined online liquid H_2S measurement using UV/Vis and electrochemical approaches. These devices may be useful downstream of pressure sewers or near dosing and treatment locations, but solids, calibration, and inspection remain relevant. A gas sensor cannot be treated as a substitute for a dissolved measurement, and a generic battery or replacement interval should not be assumed [Despot et al., 2021, Pandey et al., 2012].

4.6 Condition and defect monitoring

Acoustic sensing can inspect pipes for blockage or defect signatures without requiring intrusive manhole entry; CCTV directly reveals structural and operational defects; computer vision automates defect classification when training libraries exist; and distributed fibre-optic sensing can profile thermal anomalies or strain. These technologies provide complementary diagnostic evidence that continuous level monitoring cannot capture on its own: a sewer pipe may display normal dry-weather hydraulic depth while harbouring severe structural fractures, root balls, sediment accumulation, or encrustation.

Recent experimental research in South Australia by Ninh et al. [2026] demonstrates the proactive capability of acoustic monitoring using paired acoustic sensors. The system positions an acoustic transmitter and receiver in adjacent maintenance holes, transmitting continuous acoustic waves through the sewer pipe headspace. By analyzing acoustic attenuation and dispersion across multiple frequency bands, the paired-sensor system successfully detected and quantified cross-sectional pipe blockages ranging from 10% to 60% obstruction under varying dry-weather flow depths. Crucially, acoustic attenuation identifies physical constriction during normal, low-depth flow conditions long before a hydraulic surcharge occurs. This contrasts with non-contact level sensors, which only detect a blockage after sewage backs up and rises upstream. Paired acoustic sensing thus offers a rapid, non-destructive screening layer capable of rating pipe condition and pre-screening candidate manholes ahead of fixed level sensor deployment [Ninh et al., 2026].

4.7 Telemetry, power, and maintenance

Telemetry is part of sensor placement rather than an afterthought. LoRaWAN, cellular, gateways, and platform integration determine whether a measurement can be used in time. Sampling more frequently increases energy and data demand. Access, cleaning, calibration, battery replacement, corrosion, and safe maintenance can dominate the purchase price. Non-contact installation reduces immersion exposure but does not remove condensation, surface fouling, or mounting problems. The final deployment objective should therefore include life-cycle cost and maintainability, not just the number of devices.

Sydney Water's IoT programme puts numbers on that trade-off at utility scale. Devices are specified for the sewer access chamber rather than for the sensing task alone: casing that withstands a highly corrosive environment with no aluminium or steel, IP68 for high humidity and occasional immersion, cabling and seals resistant to vermin and cockroaches, mounting brackets designed for safe and efficient installation, and external antennas at approximately 90% of sites because the devices sit in concrete pits under metal lids. NB-IoT is preferred over LoRaWAN for this metropolitan use case. Battery life is treated as a first-order design driver, and the consequence is a telemetry cadence rather than a sensing limit: ultrasonic and radar level devices measure every 15 minutes and submit all readings once per day, so a backend comparison between two instrumented sites can only be performed retrospectively over the previous 24 hours and a real-time alarm based on the difference between two individual monitoring sites is not currently possible. Call-in frequency can be raised, at the cost of battery consumption. This is a direct constraint on any placement method whose objective assumes synchronised, high-frequency observation of adjacent chambers, and it should be checked against SA Water's own telemetry standard before such an objective is adopted [Drenoyanis and Prackwieser, 2022].

In utility engineering practice, candidate chamber selection must also conform to strict physical and hydraulic siting standards. The American Society of Civil Engineers' Manual of Practice No. 60 (MOP 60) establishes that sewer monitoring instrumentation must be sited in reaches maintaining undisturbed, non-surcharged gravity flow, requiring a straight upstream run of 5 to 10 pipe diameters free from hydraulic drops, confluences, or backwater regimes [ASCE and WEF, 2007]. Siting must also avoid hazardous confined-space entry, ensuring surface-accessible mounting brackets. These criteria are reinforced by UK Water Industry Research (UKWIR) guidelines and United Utilities' Sefton Dynamic Network Management deployment (20,000 sensors), which demonstrate that device longevity is governed primarily by battery drain from cellular NB-IoT penetration through heavy cast-iron chamber covers, necessitating automated watchdog diagnostics and adaptive reporting cadences [UK Water Industry Research, 2021, United Utilities, 2022].

A recent smart-sewer review also treats instrumentation, applications, and pathways to adoption as one implementation problem. This supports keeping deployment, data quality, maintenance, and organisational use in the same design conversation as the sensing technology [Rousso and Gong, 2026].

5 Sensor-placement and optimisation methods

5.1 Method families

The review of urban drainage sensor arrangement groups methods into five broad families and shows why no single family is sufficient for every network [Ma et al., 2025b]. Three further families, added below, cover the metaheuristic search, Bayesian design, and GNN-based methods surveyed elsewhere in this section but not part of that original taxonomy.

Table 5: Comparison of sensor-placement method families.

Family	Typical data	Strength	Weakness
Engineering experience or rules	Asset knowledge, known hotspots, expert judgement.	Fast, practical, and usable when network data are missing.	Does not quantify hydraulic or time-series information consistently and can break down on large, complex networks.
Information theory	Time series, entropy, mutual information, joint dependence, redundancy.	Quantifies information and can distinguish unique from duplicated measurements.	Computationally heavier and sensitive to missing or old time series, normalisation, noise, and the chosen objective.
Cluster analysis	Feature vectors, hydraulic states, spatial or temporal similarity.	Can reduce a large set of similar candidates and support representative sampling.	A cluster does not necessarily identify the best point within the cluster; results depend on features and distance.
Complex network theory	Directed topology, centrality, node relevance.	Fast first-pass screening without a hydraulic simulation.	Can overlook flow, decay, storage, capacity, and threshold behaviour.
Observability and reconstruction	Hydraulic model, state equations, measurement matrix.	Connects locations to the ability to estimate unmonitored states.	Requires a model and can degrade with sparse, noisy, nonlinear, or poorly placed measurements.
Added beyond Ma et al. [2025b]'s taxonomy:			
Metaheuristic/exact search (GA, NSGA-II, greedy, tabu, MIP)	Candidate set, explicit objective(s), budget.	Optimises directly against a stated, often multi-objective goal.	No optimality guarantee (metaheuristics) or poor scaling (exact solvers).
Bayesian experimental design	Prior/posterior distributions, forward model or surrogate.	Makes the placement objective explicit; can guide calibration jointly.	Many forward-model evaluations needed; often requires a surrogate.
GNN / deep-learning reconstruction	Monitored time series, topology, optional pipe attributes.	Learns reconstruction from data without an explicit hydraulic model.	Needs per-catchment training data; transferability largely untested.

The same review found no standard evaluation protocol. Algorithm metrics include runtime, hypervolume, silhouette score, and Calinski-Harabasz score; outcome metrics include RMSE, R^2, NSE, detection rate, response time, reliability, uniformity, and resilience. The team should choose outcome metrics that match the operational question rather than reporting an algorithm score alone [Ma et al., 2025b].

5.2 Minimum information and minimum redundancy

Li, Singh, and Mishra proposed a minimum-information, minimum-redundancy (MIMR) criterion using joint entropy, transinformation, and total correlation. The purpose is to select stations that collectively contain high information while avoiding repeated information.

In a 12-gauge Brazos River case, information was not distributed simply downstream. The MIMR top six contained less than 20% duplicated information, compared with 32–72% for the water-management-plan layout, which was more clustered. In the 1,520-candidate Delfland case, 16 near-pumping-station candidates had zero information, the top nine stations captured about 85% of the available information, and the tenth added little. MIMR again spread stations more evenly and reduced redundancy. Weighting values between 0.5 and 1.0 produced stable rankings in the tested cases, but that stability may not hold in a larger or denser network.

The method needs dense candidate data and time series, does not directly include economic cost, and is more computationally involved than simple matrix-based methods. Its project value is the explicit treatment of redundancy: two sensors that see almost the same hydraulic pattern may offer less value than two sensors with slightly lower individual information but different failure or flow contexts [Li et al., 2012].

5.3 Mutual information and modified greedy selection

Crowley et al. selected sewer sensor locations by maximising mutual information between network states and sensor observations. Under the paper's Gaussian-state and Gaussian-noise assumptions, the mutual information can be expressed through the determinant of covariance-related matrices. The objective is nondecreasing and submodular under the stated conditions, which gives the standard greedy algorithm an approximately 63%

lower-bound guarantee relative to the optimum. The study used a one-step modified greedy search: standard greedy selection was run for every possible initial sensor and the best sequence was retained.

Two real networks were used. The south England network contained 479 nodes and 567 links, served about 12,000 people over approximately 5 km^2, and had two dry-weather days of flow at two-minute resolution. The Bellinge, Denmark network contained about 1,020 nodes and about 1,000 links, with two years of SWMM flow and depth at two-minute resolution plus rainfall. The selected layouts were compared with random placement, a maximum-state-sum heuristic, and rule-based locations near outfalls, large subcatchments, or trunks. Generalised linear models and generalised regression neural networks then estimated unmonitored states, and normalised mean squared error (NMSE) assessed reconstruction.

In the project deck's summary of the study, the selected layouts delivered approximately 40% mutual-information gain at 5–10 sensors and at least 30% at 25–50 sensors relative to comparison placements. The source study also found improvements over random and simple heuristics in many sensor-count cases, with exceptions depending on network and predictor. The method tends to select clustered locations, may treat all nodes equally when consequence weights are not added, and needs a hydraulic model, GIS, compatible measurement units, and enough time-series data. It is therefore a strong candidate for flow or state reconstruction after the client data are obtained, but it must be modified if environmental risk or high-consequence outfalls are mandatory constraints [Crowley et al., 2025, PG S2-22 project team, 2026c].

5.4 Bayesian optimal experimental design and goal-oriented information gain

Cheng et al. address a question that sits underneath every method in this section: what the placement is supposed to be optimal *for*. Bayesian optimal experimental design assigns each candidate location an expected utility over prospective observations. When that utility is the Kullback–Leibler divergence from the posterior to the prior, the resulting expected information gain (EIG) measures how strongly a prospective observation should change the distribution of the learning target, and is equivalently the mutual information between the observation and that target. This is the same quantity maximised in Section 5.3, but with the target declared explicitly rather than assumed. The paper contrasts parameter-oriented design, which values EIG about the model parameters, with goal-oriented design, which values EIG about a stated prediction obtained by pushing the parameter distribution through a fixed map. A- and D-optimality are noted as simpler covariance-based alternatives.

The case is the August 2014 storm in the I-696 to Mound Road corridor of metropolitan Detroit, with 2,576 candidate locations and a six-parameter uncertainty model. Flood physics came from tRIBS-Urban, representing distributed hydrologic processes and two-dimensional overland flow; because the design calculation requires many model evaluations across parameter realisations and candidates, a deep neural-network surrogate was substituted and global sensitivity analysis run alongside the EIG estimates. Targets included point depths one hour after observation, their joint vector, regional-average depths, and vectors of regional maximum depth at three lead times.

Three findings are relevant here. Learning the full parameter vector transfers unevenly into reductions in predictive uncertainty across locations and lead times, so a parameter-oriented design is not automatically a good prediction-oriented design. The prediction target changes the spatial ranking: point-depth targets favoured nearby locations, while regional-average and regional-maximum targets could favour non-local ones, and weighted multi-point objectives preserved broad spatial patterns while shifting the computed maximum. Information value and deployability require different evidence and belong in the same process: public geospatial classifications can organise an initial screen, but only verified field conditions determine which informative locations can actually be installed and operated, since access, safety, power, communications, permission, maintenance and organisational responsibility all restrict placement.

The recommended practice is an objective-first workflow: define the intended prediction and any priority weights, apply field-verified restrictions to fix the admissible set, compute the information map, identify high-EIG locations, then interpret them with hydraulic and graded feasibility evidence. The transfer limits are clear and stated by the authors: one urban domain, one historical precipitation forcing, a single-sensor design problem, a six-parameter uncertainty model, and a surrogate not yet validated across wet and dry regimes or at high-EIG locations. The case is also urban pluvial flooding rather than in-sewer monitoring, so the contribution to this project is the design principle and the feasibility screen, not the numerical rankings [Cheng et al., 2026].

5.5 Complex network centrality and hydraulic reliability

Simone et al. compared a deterministic global optimisation method with a topology-based in-relevance harmonic centrality method for contaminant monitoring. The global method used SWMM to simulate transport and decay, binary sensor decisions, concentration thresholds, and the SCIP mixed-integer global solver. A pollutant was inserted at one node at a time; reliability was the proportion of injection points detected above threshold by at least one sensor.

In the 77-node, 79-link benchmark network, injections were 1 mg/L and first-order decay was 0.2 h^{-1}. At a threshold of 0.008 mg/L, four sensors achieved reliability 1 in the reported case, and the solver completed in less than one second. In the larger 531-node, 530-pipe real case, full reliability required four sensors at a 0.003 mg/L threshold, eight at 0.008 mg/L, and eleven at 0.01 mg/L. This is a contaminant-detection result under a specified hydraulic and decay model, not a general sewer coverage guarantee.

The in-relevance harmonic centrality method uses directed topology and node relevance such as inflow or contaminant quantity. Outfalls and main-path nodes were highly ranked, while head nodes had little centrality. Centrality locations were close to the optimisation locations in the tested cases and can reduce the search space before hydraulic optimisation. Its limitation is equally important: topology alone cannot represent dilution, storage, threshold changes, or hydraulic model error. It is best used as a transparent preliminary screen or as one objective in a hybrid method [Simone et al., 2023].

5.6 Multi-objective cost-coverage-resolution optimisation

Wang, Xu, and Zhou formulated a directed sewer network placement problem for pandemic monitoring using three objectives:

1. minimise sensing cost;
2. maximise upstream monitoring coverage; and
3. minimise the expected manual search cost when a signal is found.

The coverage and search objectives were calculated from directed connectivity, upstream manhole priority, and overlap. If many nodes are covered by one downstream sensor, a positive signal may still require a large manual search. The paper called this cost-coverage-resolution (CCR) optimisation. An NSGA-II genetic algorithm used an initial population, non-dominated sorting, crowding distance, single-point crossover, bit-string mutation, and a stopping rule of 50 iterations without significant change.

The Tuen Mun case contained 4,394 nodes, 4,308 edges, and more than 500,000 residents. The optimised solutions used a restricted set of 236 manholes, about 5.3% of the network candidates. About 20 sensors provided around 80% coverage in the reported trade-off set, while about 80 sensors approached 90%. At 100 sensors, an optimised layout achieved 80.9% coverage against 64.5% for a human-heuristic initial layout. The final hypervolume was 0.550 against 0.271 for the strongest human heuristic, described as a 102% improvement. These results demonstrate the value of global multi-objective search and the inclusion of search effort, but the study did not include SA Wateraccess, flow, asset condition, or field trials [Wang et al., 2024b].

5.7 Evolutionary greedy scaling of the same objective

Wang, Xia, Chen, Tong, and Zhou revisit the cost-coverage-resolution problem above with an Evolutionary Greedy (EG) algorithm built for computational efficiency on large, directed sewage networks rather than for a marginally better trade-off curve. Benchmarked on synthetic small networks and the same Hong Kong network, EG matches the solution quality of the NSGA-II search while running substantially faster – relevant to the project only once a candidate set approaches city scale, but a useful fallback if the multi-objective search above proves too slow to iterate on [Wang et al., 2024a].

5.8 Machine learning for contaminant source identification and placement

Salem and Abokifa linked sensor placement to the ability to identify a contaminant source. The nested method first used a multilayer perceptron to forecast concentrations at possible sensor locations, then used a genetic algorithm to optimise the placement. The objective combines source observability and source-identification reliability:

$$S_{\text{tot}} = I_{\text{obs}} S_{\text{obs}} + I_{\text{rel}} S_{\text{rel}},$$

where the weights express the decision-maker's preference.

The study trained on 150 Latin-hypercube events. Each event used two simultaneous sources selected from 53 drainage junctions, with concentrations from 1 to 100 mg/L. PySWMM generated tracer concentrations; binary connectivity and distance matrices represented sensor reach and source distance. The medium combined sewer case contained 84 conduits, 80 junctions, 59 residential sheds, an area of 308,350 m^2, and a dry-weather peak of 42.8 m^3/h.

The results showed a direct trade-off: downstream sensors covered a larger area and improved observability but diluted the signal, while upstream sensors preserved source reliability but saw a smaller area. A two-sensor high-decay case selected a combination of upstream and downstream locations. For a low-decay case, middle-network locations better preserved source uniqueness. Adding a third sensor to two existing sensors improved reliability by 16% but observability by only 1.8%. Weighting changes the selected locations, and additional

sensors improve reliability more than observability in the tested cases. The limitations are the SWMM first-order decay and single-species assumptions, simulated training data, and the need for a more advanced water-quality model or Bayesian uncertainty treatment for a different network [Salem and Abokifa, 2024].

5.9 Topology- and elevation-derived blockage detection

Ninh, Do, Zeng, and Lambert combine a genetic algorithm with a topology- and elevation-derived *blockage detection condition* to size and lay out sensors across two real gravity sanitary sewer systems over multiple investment phases, targeting blockage and overflow detection directly. A candidate location s is only counted as covering a downstream node if

$$Z_0 + H_d > Z_s + H_s,$$

where Z_0 and H_d are the invert elevation and overflow-triggering depth threshold (70% of full pipe depth) at the sensor's downstream node, and Z_s, H_s are the invert elevation and typical dry-weather depth at the candidate location itself. This makes coverage *position-dependent*: the same sensor covers a different set of nodes depending on where it sits in the network, rather than a fixed radius or a uniform detection assumption as in most of the methods above and below. The authors report coverage capacity varying substantially between manholes even within the same system, and the method explicitly supports multi-phase investment scheduling – a capability few other reviewed methods offer. For the project, this is the closest published precedent for scoring a candidate location by *how much of the network it can actually see from there*, not just by its topological position [Ninh et al., 2025].

5.10 Bayesian placement with joint model calibration

Huang, Xiong, Zhang, Zheng, Ji, and Gupta place sensors and calibrate the hydraulic model together, on the practical grounds that a model is hardest to calibrate exactly when it is least observed. A graph-partition step spreads candidate sensors spatially; Bayesian Experimental Design then selects locations by the expected relative entropy between prior and posterior parameter distributions, before a sequential ensemble-smoother calibration step runs. On two real urban drainage models under varying rainfall and parameter scenarios, the joint approach outperformed even spatial distribution and flood-hotspot placement on both prediction accuracy and uncertainty reduction, though it assumes a uniform Gaussian sensor observation-error model at every candidate location rather than the position-dependent detection capability of Ninh et al. [2025] above [Huang et al., 2025].

5.11 Value of information, transinformation, and matrix completion

Zheng et al. combined information-based placement with reconstruction-based evaluation in an 878-node urban drainage network. Value of information (VOI) and transinformation entropy (TE) selected locations; non-negative matrix factorisation (NMF) evaluated whether sparse observations could reconstruct system states. Four sensors produced 76% flood-classification accuracy and 33% operation-status error; eight sensors produced 82% accuracy and 29% error. The main improvement occurred between four and eight sensors, after which the curve flattened up to 20 sensors.

The study found that VOI rose approximately linearly with sensor count while redundancy did not increase at the same rate. Sensor value depended on the flood probability at the node and was highest near a probability of 0.5, not on physical node distance. An optimised three-sensor arrangement outperformed the existing three-sensor arrangement and had roughly two orders of magnitude lower redundancy. A river section was simplified or excluded from the placement domain because matrix completion performed poorly there; the implication is that receiving-water or river sections may need a separate model. The cost values used in the analysis were illustrative, so real SA WaterCapEx and OpEx would be required before using the objective [Zheng et al., 2025].

5.12 Risk-driven machine learning and weighted set cover

Moradi and Najafi developed a data-driven pipe-prioritisation and sensor-placement framework called DPP-SP. The method first predicts pipe failure risk from static network and asset attributes, then selects sensor locations to maximise weighted upstream coverage within a detection radius.

The anonymised US county case contained approximately 32,039 gravity pipes, 1,310 force-main pipes, 31,865 manholes, 34 pumps, 77 large meter vaults, 127 sewer vaults, and 48 current monitoring stations. Failure labels came from 2020–2025 CCTV and ITpipes records, service-disruption logs, and sanitary-sewer-overflow history. Of the network, 22,130 pipes, about 66%, had CCTV observations. A positive label was assigned for a maximum PACP condition grade at or above 3, a maintenance issue, or an SSO, producing 485 positive labels out of 33,474 reconstructed network elements, or about 1.454%.

The network reconstruction integrated 200 vault nodes, inserted 1,407 virtual junctions, re-snapped 33,349 endpoints, and merged 28 duplicate pairs. The final graph had 33,474 nodes, 33,349 edges, and 78 components;

the largest component contained 29,914 nodes, 89.4% of the graph. Six hundred and eighty candidate locations were drawn from existing sites, pumps, vaults, and high-confluence manholes. Reverse-Dijkstra upstream reachability used a 1,500 m maximum detection distance, corresponding to about 21–42 minutes at assumed velocities of 0.6–1.2 m/s.

The prediction features were diameter, age, slope, length, upstream degree, downstream degree, material, and pipe class. A stratified 75/25 split compared a 300-tree balanced random forest, XGBoost, and an MLP with 128 and 64 hidden units, ReLU, and 0.3 dropout. On the holdout set, random forest ROC-AUC was 0.7626, PR-AUC 0.0844, and F1 0.1684; XGBoost was 0.7323, 0.0654, and 0.1134; MLP was 0.6708, 0.0800, and 0.1806. Repeated stratified five-fold cross-validation also favoured random forest, with mean ROC-AUC 0.750 and a 95% interval of approximately 0.741–0.758.

The placement risk was $r_i = p_i s_i$, where p_i was predicted probability and severity s_i was 1 below 203 mm, 2 from 203–457 mm, and 3 at or above 457 mm or for force mains. Against a fixed reference total of 11,311.44 risk units, the 48 existing stations covered 11.73% or 1,327.19 risk units and 3,355 assets. The random-forest greedy layout covered 32.26% or 3,649.11 risk units and 8,187 assets, a 174.9% relative improvement over the existing layout. Genetic algorithm and tabu-search layouts were within 0.01 percentage points of greedy; greedy ran in less than one second, GA in about 8–11 seconds, and tabu search in about 87–117 seconds.

Coverage increased with more sensors but plateaued: 32.26% at 48 sensors, 40.12% at 78, 43.87% at 100, 49.54% at 150, 54.57% at 250, and 56.58% at 400. The candidate and radius design left 77 isolated fragments and about 43% of risk unobservable. Sensitivity to detection radius was large: at 48 sensors, coverage ranged from 4.22% at 200 m to 43.30% at 2,000 m; at the final 400-sensor design, the 1,500 m ceiling was 56.58% and a 2,500 m radius reached 63.31%. Age and length were the dominant predictors; permutation importance reduced ROC-AUC by 0.103 for age, with SHAP means of 0.111 for age, 0.103 for length, 0.070 for slope, and 0.050 for material.

The framework is highly relevant to the project because it connects failure probability, consequence, reachability, candidate density, and existing assets. It is not a SA Waterresult: the case was anonymised, used static attributes and no dynamic hydraulics, and produced a coverage ceiling specific to its candidate and radius assumptions [Moradi and Najafi, 2026].

5.13 Topology-derived flow inference for inflow and infiltration detection

Wang, Du, Zheng, Liang, Lu, Kapelan, and Savic address inflow and infiltration (I&I) detection specifically, arguing that fixed, isolated detection thresholds cannot capture a genuinely heterogeneous flow pattern – their own case network has one sub-area covering 16.7% of the service area that contributes roughly 32% of daily flow. Their topology-derived flow inference (TDFI) method synthesises information from multiple upstream sensors into a location-specific *Individual Pipe Detection Threshold* (IPDT), chosen via a deterministic Sequential Backward Selection algorithm. Critically, an unmonitored pipe's IPDT is capped at its own full-pipe hydraulic capacity from the Manning equation,

$$Q = \frac{D^{8/3} I^{1/2}}{4^{5/3} n},$$

using diameter D, slope I, and roughness n – the same attributes carried by the project's own GIS pipe data (diameter and grade are populated fields in the Walkerville extract; roughness is not). Prioritising sensor density in high-building-density areas materially improved I&I detection, and the method scaled to large real networks. Together with Ninh et al. [2025] above, this is the closest published precedent for a placement rule where detection capability is tied to both network position *and* pipe hydraulic capacity, rather than assumed uniform – directly relevant if the project weights candidate locations by exposure to capacity exhaustion rather than by failure risk alone [Wang et al., 2026].

5.14 GIS-based fuzzy AHP suitability screening

Nguyen, Bui, and Seidu take a non-algorithmic, multi-criteria route: a GIS-based fuzzy Analytical Hierarchy Process (AHP) weights eight criteria – altitude, rainfall, geology, manhole locations, population density, critical-infrastructure proximity, road network, and traffic load – to produce a flood-sensor *suitability map* for the wastewater/stormwater network of Ålesund, Norway, instead of optimising an explicit coverage or information objective. Manholes, altitude, and rainfall were the most sensitive factors. The method does not compete with the optimisation approaches above; it is a candidate-screening layer that could sit ahead of any of them, using several of the same geographic and asset attributes already in the project's own scope [Nguyen et al., 2020].

5.15 Risk-based site selection in utility practice

Drenoyanis and Prackwieser describe how a large Australian utility actually chooses monitoring sites, which provides a practitioner baseline against which the computed methods above can be compared. Sydney Water

reports that most blockages occur in small-diameter reticulation pipes and are caused by tree roots penetrating pipes and access chambers, wet wipes, fat, oil and grease, or a combination. Site selection follows a ten-step process: prioritise a sub-catchment by number of dry-weather overflows and licence limits; inspect the asset on site including communication-signal testing; carry out repairs and root-cutting; assign each asset a risk rating combining the likelihood of blocking or choking with the environmental consequence of failure, incorporating sewer location and pipe size; generate a risk rating for pipeline chains by calculating the probability of failure of assets connected in series; produce an asset priority list; have subject-matter experts assess sites spatially; assess safety and accessibility; run environmental, heritage and traffic-control assessments where required; and refine the delivery plan. The chain-of-assets step is a coverage argument in engineering form and is the part most directly comparable to the weighted set cover of the preceding subsection.

Three detection methods are reported with their limits. Comparing wet-well inflow rates and pump runtimes against historical diurnal patterns at a pumping station can detect overflows and some blockages anywhere upstream, but it depends on the lost flow being statistically significant against normal station inflow, so chokes that are a small proportion of typical flow cannot be separated from natural variation, and the fault is localised only to what may be tens of kilometres of network. Instrumenting reticulation access chambers reduces that search area and allows a two-sided diagnosis: a sustained normalised positive flow error indicates a choke downstream of the chamber and a sustained negative error one upstream. Correlating consumption from upstream digital water meters against derived sewer flow gave a correlation of 0.97 on a dry day but 0.79 on a rainy day, and water used for irrigation and pool top-ups may never enter the sewer at all.

The reported outcome is an average of 20 blockages detected per month, with more than 300 blockages at environmentally high-risk sites identified and cleared, representing approximately $400,000 per month in avoided incident costs, which the authors state is outweighed by non-financial benefits including environmental protection, community amenity, public health and employee safety. Two operational observations are worth carrying into the project's objective function. Level sensors are preferred over float switches because they show how fast the level is rising, which combined with historical flow and time of day is used to estimate required response times. And many of the chokes found so far were in bushland, where an overflow is not in public view and could run for days before being reported — a consequence-of-failure argument for instrumenting where nobody is watching, which is the opposite of what an unweighted coverage objective would produce. The transfer limits are that this is an industry operational report reporting no detection rate against an exhaustive ground-truth blockage set and no false-alarm statistics, and that the figures are specific to Sydney Water's network, licence conditions and cost base [Drenoyanis and Prackwieser, 2022].

Broader Australasian and international utility grey literature reinforces this risk-weighted operational approach while highlighting how smart monitoring substitutes for capital augmentation. Under the Water Services Association of Australia (WSAA) Wet Weather Overflows series, Sydney Water collaborated with the NSW EPA under Pollution Reduction Program (PRP) 307 to replace crude, uniform spill-frequency caps with receptor-weighted environmental risk scores [WSAA and Sydney Water, 2019]. Rather than treating all overflow points equally, each discharge point is scored by multiplying spill volume by the receiving environment's ecological vulnerability, human contact potential, and foot traffic derived from anonymised mobile-phone density data. This establishes a clear regulatory precedent for the weighted set-cover objectives evaluated in this project.

In the Australian Capital Territory, Icon Water faced severe capacity bottlenecks in Canberra driven by incremental urban densification that outpaced trunk pipe capacity [WSAA and Icon Water, 2019]. Instead of executing traditional, disruptive pipe duplications estimated at A$173M, Icon Water implemented a multi-criteria risk framework that accepted smart level monitoring, real-time alarms, and proactive maintenance as regulatory-approved alternatives to capital upsizing. Similarly, Melbourne Water's regional trunk standard operates on a 1-in-5-year average recurrence interval (ARI) containment philosophy, relying on upstream retention attenuation and dynamic SCADA control to defer capital works while demonstrating that wet-weather sewage overflows contribute less than 1% of the annual contaminant budget to receiving bays [WSAA and Melbourne Water, 2019]. Across the UK water sector, UK Water Industry Research (UKWIR) evaluated heuristic versus algorithmic sensor placement across 10 water companies, concluding that purely algorithmic coverage must be bounded by utility accessibility and historical blockage hot spots [UK Water Industry Research, 2021]. This aligns with the baseline risk and vulnerability assessment (BRAVA) framework adopted across England and Wales under statutory Drainage and Wastewater Management Plans (DWMP), which standardizes risk scoring across asset vulnerability, hydraulic surcharge probability, and receptor impact [Water UK, 2018].

5.16 Digital-twin sparse sensing with SVD and QR factorisation

Ding et al. proposed data-driven sparse sensing (DSS) for flow reconstruction. The approach uses EPA-SWMM to generate a full spatiotemporal flow matrix, singular value decomposition (SVD) to identify a reduced modal basis, and QR factorisation with column pivoting to select representative sensor locations. Measurements at those locations reconstruct the wider flow field in the reduced basis.

The Woodland catchment in Duluth, Minnesota covered 133.53 ha, served approximately 6,244 residents, and

contained 48 subcatchments, 130 conduits, and 77 nodes. The training ensemble combined ten rainfall events, including five design storms and five observed storms, with 25 hydrologic parameter settings: five imperviousness levels and five Horton minimum infiltration rates, producing 250 scenarios. Six events were held out for testing, including five observed storms and a 200-year design storm. The SWMM model was physically grounded but not validated against field measurements, so it was used as a scenario generator rather than a calibrated operational digital twin.

NSE measured reconstruction agreement. Exhaustive search was used for one to four sensors and Monte Carlo sampling of 20,000 configurations per count for five to ten sensors. The mean system-level NSE for observed events rose from 0.896 with one sensor to 0.993 with ten; the 200-year event rose from 0.842 to 0.994. Three sensors achieved mean observed-event NSE 0.949. The DSS layouts were close to exhaustive optima for one to four sensors and outperformed random layouts for five to ten, with mean NSE from 0.983 to 0.995 in that range. At six sensors, 5%, 10%, and 15% multiplicative Gaussian noise left median NSE at approximately 0.981 ± 0.001.

Failure testing showed that most single-sensor dropouts had limited effect, but a few locations were critical. At six sensors, removing three tested locations produced mean NSE losses of 0.009, 0.038, and 0.007; at ten sensors, most losses were below 0.055 but two locations caused losses of 0.299 and 0.367. The mean loss fell from 0.476 at two sensors to about 0.358 at six and 0.367 at ten, while worst-case sensitivity remained high. The method was robust in average reconstruction across SWMM parameter variants, but exact sensor locations changed with model parameterisation. The approach is transferable as a reconstruction method, but the case is stormwater rather than wastewater and contains no SA Waterfield validation [Ding et al., 2025].

5.17 Graph neural network based state reconstruction and placement

A growing cluster of research reframes placement around data-driven reconstruction: rather than only choosing locations, a graph-aware model learns to infer hydraulic state at every *unmonitored* node from a sparse set of monitored ones. He, Nan, Chen, Ye, Ji, Chen, Zhang, Wu, Liu, Ge, and Che's spatiotemporal graph convolutional network (STGCN) reconstructed water levels at 527 unmonitored nodes from only 7 monitoring nodes at 71.3% accuracy [He et al., 2025b]. A follow-up edge-attribute-enhanced STGCN added physical pipe attributes – notably invert elevation – as edge features, lifting reliable predictions to 88.6% of nodes at only 10% monitoring density, well above a plain MLP (14.5%) or a topology-only STGCN (47.0%) [He et al., 2025a]. He, Nan, Chen, and Ye then extended this into an explicit placement framework built on an *ideal perceptual domain* (IPD) metric that estimates the theoretical coverage a candidate sensor set could achieve before any reconstruction model is trained, checked against an *actual perceptual domain* (APD) achieved by a graph-based reconstruction model; targeting 90% IPD coverage needed sensors at only 3.0% of network nodes and achieved 86.7% APD coverage in a representative rainfall event. The authors note, however, that pipe diameter and other physical attributes feed only their reconstruction model and are *not* used in the placement step itself – an acknowledged gap between the capacity-relevant data these GNN studies already extract and its use in choosing sensor locations [He et al., 2026].

5.18 Unsupervised clustering and forecast-based placement

Two lighter-weight, unsupervised approaches address the same class of problem. Qi, Li, Cao, Li, Zhang, and Yang combine a graph convolutional network for topological feature extraction with a structured deep clustering network to identify critical monitoring zones without labelled training data, achieving over 94% agreement with expert-defined labels and cutting required monitoring points by more than 40% while increasing critical-node coverage above 90% on the public C-Town benchmark [Qi et al., 2025]. Chen and Yu target Low Impact Development (LID) stormwater systems with a cluster-then-forecast framework: a clustering step identifies representative centroid nodes for sensor placement, and an LSTM then forecasts system-wide water levels five minutes ahead from the preceding 25 minutes of monitored data, addressing flood hotspots that shift under nonstationary rainfall and urbanisation [Chen and Yu, 2025].

5.19 Review evidence on algorithms and AI/ML

The urban wastewater review screened 189 Web of Science results and retained 96 studies through December 2024. It separates objectives into operational-state monitoring (flow, level, velocity, model calibration) and abnormal-state monitoring (pollutant tracing, source identification, wastewater epidemiology, flood or overflow hotspots). It also identifies a recurring assumption: equal likelihood at every node. Weighted priors based on population, downstream position, or known risk can reduce computation and better represent the actual decision.

Representative cases in the review include entropy selection in a 1,909-node network using 12–14 points; reduction from 748 candidates to 20; four and eight sensors in the 878-node VOI/TE case; four to seven sensors in a 77–79-node complex-network study; four to ten sensors in a 647-node waterway; clustering from 23 to 10 points; and a 62.88 km^2 area in which clustering reduced seven locations. Other cases selected 20 points from

819–820-node networks for infiltration and inflow analysis. The review concludes that hybrid methods and phased deployment with feedback are often more useful than a one-time universal optimisation [Ma et al., 2025b].

Moradi et al. describe exact, linear, convex, greedy, genetic, metaheuristic, gradient-based, GRASP, tabu, and simulated-annealing approaches. Exact and convex methods can be accurate but do not scale easily; greedy methods are efficient but can be locally suboptimal; evolutionary methods explore more alternatives but cost more computation; gradient methods can suit large problems but depend on representation and hyperparameters. The review also maps AI/ML roles: decision trees for interpretable rules, random forests for high-dimensional feature selection and importance, logistic models for simple relationships, SVMs for high-dimensional classification, kNN for smaller feature spaces, Gaussian processes for uncertainty, and neural networks for complex nonlinear relationships. The project should choose a model because its inputs and validation match the operational question, not because it is labelled AI [Moradi et al., 2025].

6 Sewer condition, defects, and failure factors

6.1 Factor catalogue from the systematic review

Mohammadi et al. reviewed 23 sewer condition-prediction studies published between 2001 and 2019 and compared condition-rating standards including WRc, PACP, and WSAA-related approaches [Malek Mohammadi et al., 2020]. The literature consistently identifies pipe age, material, diameter, and length, although the effect of diameter is contradictory across studies. Physical attributes are usually cheaper and more available than environmental and operational attributes.

Operational guidance from the US Environmental Protection Agency's Capacity, Management, Operation, and Maintenance (CMOM) programme establishes a critical operational distinction: across municipal collection systems, pipe blockages account for roughly 48% of all overflow incidents, whereas hydraulic capacity limitations and peak wet-weather flows account for nearly 75% of total overflow discharge volume [United States Environmental Protection Agency, 2005]. This dichotomy reveals why monitoring objectives cannot be uniform: small-diameter reticulation pipes fail primarily through physical chokes (roots, grease, debris), requiring sensitivity to localized level rises, whereas large trunk mains fail through capacity exhaustion during storm events, requiring dynamic flow and hydraulic headroom tracking.

Table 6: Condition and failure factors relevant to monitoring priority.

Factor group	Examples	Placement implication
Physical and geometric	Start and end invert, installation method, joint type, length, shape, slope, age, material, size, depth.	Identify pipes where deterioration or hydraulic consequence may be higher; support graph, grade, capacity, and severity features.
Environmental and ground	Backfill and bedding, ground movement, groundwater, pH, road type, roots, soil corrosivity, fracture, moisture, type, sulfate, traffic.	Weight sites where soil, roots, traffic, or groundwater increase failure likelihood or where access is difficult.
Operational and hydraulic	Blockage, burst history, debris, flow velocity, hydraulic condition, infiltration or exfiltration, previous maintenance, sediment, sewer type.	Use incidents and hydraulic anomalies to prioritise locations that need early warning or condition inspection.
Condition evidence	CCTV defect type and frequency, structural and operational grades, inspection date, repair history.	Convert inspection evidence into labels or risk weights; avoid treating a live level reading as a condition label.

6.2 Bayesian network condition model

Ma et al. built a data-driven Bayesian network (BN) from a multi-source Hong Kong database. The authors first retrieved literature using sewer-condition and sewer-defect keywords, screened 19 articles, and identified 16 frequently used influence factors. They then integrated GIS pipe properties, CCTV records, census data, land use, road and traffic information, soil type and corrosion, rainfall, temperature, and humidity. CCTV defect records were linked by unique pipe ID; environmental data were joined spatially; climate data were matched to the relevant weather station and inspection date.

After duplicate inspection records were consolidated and records with excessive missing information were removed, 23,000 pipe records remained. Continuous variables were discretised and categorical states simplified. Pipe condition was represented in six levels: perfect, excellent, good, fair, poor, and critical. The learned BN contained 15 nodes and 31 links. Model interpretation used mutual information (MI), joint probability, true risk influence (TRI), scenario analysis, two probability-consistency axioms, and one historical record held out from model training.

The dataset had pipe age from 1 to 116 years with mean 37.55 years; length mean 18.35 m; diameter mean 420.23 mm; traffic mean 21,444.52 AADT; population mean 16,804.62; rainfall mean 124.15 mm; humidity mean 77.68%; and temperature mean 23.48°C. Deposits were the largest defect category at 20.19%, followed by fracture at 13.84%, displaced joint at 13.39%, and crack at 12.09%.

MI ranked pipe age first (0.124), diameter second (0.032), population third (0.027), and soil type fourth (0.024). The most critical combination in the model was age 50–60 years, diameter at or below 200 mm, population at or below 15,000, and granitic rock; the probability of critical condition increased from 17% at baseline to 72% when the highest-risk states of the four factors were combined. A historical pipe record with two medium encrustations was classified as poor by inspection and received an 85% poor-condition probability in the BN scenario. These are model-specific results from Hong Kong, but the factor relationships are useful candidates for SA Waterfeature engineering and sensitivity testing [Ma et al., 2025a].

The BN approach is valuable because it can reason forward and backward, incorporate uncertainty, update probabilities with new evidence, and work with missing variables more flexibly than a simple deterministic score. Its limitations are data integration effort, discretisation choices, region-specific relationships, condition-label standards, and the possibility that a statistically important factor is not a causal control variable. A SA Watermodel should retain local categories and validation records rather than copying Hong Kong thresholds.

6.3 Inflow and infiltration as a condition and monitoring-priority signal

Rainfall-derived inflow and infiltration increases sewage volumes, reduces treatment efficiency and can overload a system during heavy rainfall, and the I&I rate is itself treated in the literature as an indicator of the operational and structural condition of the network. That makes it a condition signal computable from flow monitoring a utility already holds, rather than from inspection. Two studies show what can be extracted and at what cost.

Wang et al. developed random forest and long short-term memory models for sewer flow prediction at two sites in Ningbo, China — a combined system and a separate system — using online flow monitoring and a tipping-bucket rain gauge of 0.2 mm accuracy, with data collected at 1-minute resolution and modelled at 5 minutes. Dry weather flow was taken as the monthly average by time of day and I&I as the difference between that and observed wet-weather flow. Both models predicted flow accurately and the LSTM outperformed the random forest, with maximum relative errors on estimated I&I volume under 7%, which the authors present as an improvement on the R-value method. Two results bear directly on monitoring design. Accuracy was highest at 5-minute resolution and fell monotonically as the interval lengthened; moving from 5 to 60 minutes cost R^2 of 0.154 and 0.222 at the first site and 0.208 and 0.225 at the second for the random forest and LSTM respectively, the sequence model losing more because it has more temporal information to lose. Read against the 15-minute measurement and daily upload cadence reported in Section 4.7, this is a direct conflict between the telemetry settings that preserve battery life and the sampling rate the analytics require, and it should be resolved explicitly rather than by default. Separately, recovery time is diagnostic: flow at the combined site returned quickly to dry weather flow after rain while the separate system took two to three days, which the authors read as evidence of mixed connections or structural defects in a system that by design should not receive stormwater. The transfer limits are that I&I here is a derived quantity rather than a measurement, so it inherits the error of the assumed dry weather pattern and can even come out negative for small rainfall, and that the models are catchment-specific with no test of transfer to an unmonitored catchment [Wang et al., 2024c].

This challenge is echoed across Australasian utility case studies. In South East Queensland, Unitywater abated 90 chronic overflow hotspots by analyzing SCADA telemetry, pump runtimes, and billing records to identify excessive wet-weather response before dispatching field crews [WSAA and Unitywater, 2019]. Comprehensive smoke testing and inspection across 70,000 lots revealed that up to 50% of peak wet-weather inflow originated from illegal downpipe connections, low gully traps, and damaged private house laterals rather than utility mains. In Sydney, the Wastewater Overflow Model synthesis report revealed that attempting to eliminate wet-weather overflows purely through end-of-pipe storage containment would require over A$18B and more than 350 years of construction [Sydney Water, 2024]. Sydney Water consequently pivoted toward catchment-wide I&I source control (estimated to be 17 times cheaper), using localized flow and level monitors to verify private-side rehabilitation. Similarly, Watercare in Auckland introduced four-tier spatial capacity mapping to pause new connection approvals in sub-catchments where private I&I exhausts remaining trunk conveyance [WSAA and Watercare, 2019].

Zhang et al. reached a comparable conclusion from pumping-station SCADA in Espoo, Finland, a network of more than 900 km with approximately 200 pumping stations, of which two were studied with their subcatchments delineated by a watershed algorithm. Because Espoo has only two rain gauge sites, rainfall was taken from the Vantaa C-band dual-polarisation radar so that every study catchment had coverage — a precedent worth noting for any Adelaide catchment, where gauge density will present the same problem. Three rainfall thresholds of 0.3, 1 and 2 mm/day split the flow record into dry and wet sets for an adaptive neuro-fuzzy inference system and a multilayer perceptron network. Both produced similar I&I patterns and the fuzzy model performed better overall, but the more useful output is a screen that needs no model at all: normalised minimum dry weather

flow above zero indicates that minimum dry weather flow exceeds what water consumption alone would produce, and therefore that infiltration is present. Both subcatchments showed it, one more strongly. The analysis also rules out a confounder, finding water consumption only about 1.2% and 0.4% below the annual average across two seasonal windows, so annual variation in consumption has little effect on sewer base flow. The authors state directly that the flow data are not yet sufficient for quantification of I&I in all situations, and that both models were insensitive to the rainfall threshold in dry weather but sensitive in wet weather because the wet-weather sample was much smaller. Together with the recovery-time signature above, this gives the project two cheap I&I indicators that could weight candidate nodes before any placement algorithm is run [Zhang et al., 2020].

7 Public wastewater data in South Australia

7.1 Survey approach

The public-data assessment searched for wastewater network data for the South Australian metropolitan utility network, using anonymous queries to public ArcGIS services and checking whether returned fields contained usable values rather than placeholders. Council-owned community wastewater schemes outside the utility network were not catalogued in the updated assessment. The report records the service address, query date, record count, field completeness, and access result so that the live services can be independently re-checked before formal use [Paul Jaison, 2026].

7.2 Updated open network sources

Table 7: Open utility network data identified in the updated assessment.

Source	Layer	Records	Use and limitation
Walkerville republication of the SA Watersewer asset register	Gravity sewer mains	1,002	Complete start and end invert fields, nominal diameter, grade, flow-direction code, and 99.5% construction year. Council-boundary coverage only; no complete maintenance-hole depth, condition, incident history, or wastewater sensor readings.
Walkerville republication	Maintenance holes; property inspection points; sewer connections; inspection openings; rising mains; merged nodes; contours	360; 2,851; 2,919; 533; 7; 3,761; 6,944	Provides mapped access points and one-metre contours. Only 10 maintenance holes have usable surface level.
Statewide Infrastructure Utilities service	Waste Water Gravity Main	201,713	Open, catchment-complete extent with invert levels at each end, nominal diameter, flow direction, material, and status. No maintenance-hole layer; nodes must be derived by snapping pipe endpoints.
Statewide Infrastructure Utilities service	Waste Water Low Pressure; Pumping; Vacuum	1,146; 6,039; 491	Provides broader wastewater asset context and non-gravity network classes; asset semantics and connectivity must be checked before combining them with the gravity graph.

The Walkerville service is the only open source identified in the survey that provides a substantial public window onto the metropolitan utility network at maintenance-hole level. Its gravity-main layer contains 1,002 records. The start-invert range is 28.15–59.15 m and end-invert range 26.96–58.33 m; nominal diameter ranges from 150 to 450 mm; construction years range from 1896 to 2017. The complete pipe-end inverts are useful for hydraulic gradients after the correct flow orientation has been established. The service also publishes maintenance holes, property inspection points, sewer connections, inspection openings, rising mains, merged nodes, and one-metre contours.

The statewide service is the important update. It is openly queryable without credentials, is not clipped to an administrative council boundary, and provides 201,713 gravity mains together with low-pressure, pumping,

and vacuum layers. It solves the public pipe-extent problem but not the candidate-location problem: no maintenance-hole layer is included, so nodes must be derived by snapping pipe endpoints and cannot carry public chamber depth.

7.3 Flow-labelled invert fields and graph direction

Querying for `START_INVE > END_INVERT` returns 1,002 of 1,002 Walkerville records. That result is not evidence that every pipe runs downhill. The labels are assigned by flow: `START_INVE` is the upstream-of-flow invert whichever geometric vertex was digitised first, and `END_INVERT` is the downstream-of-flow invert. The inequality is therefore true by definition.

The updated assessment tested three interpretations at shared chambers against independent publishers:

Table 8: Agreement of invert-field interpretations at shared chambers.

Interpretation	Utility network	Statewide layer
Inverts labelled by geometry	35.2% exact	32.6% exact
Direction inferred by comparing inverts	37.0% exact	36.4% exact
Inverts labelled by flow	95.3% exact	85.0% exact

The flow-labelled reading agrees to a mean of 4 mm in the utility sample. The alternative readings disagree by approximately 0.79 and 0.85 m. Reversing the interpretation silently reverses 583 of the 1,002 utility mains and 1,867 of 3,336 tested statewide mains; record, node, and edge counts remain unchanged. Under the correct reading, inconsistent chamber inverts fall from 462 nodes to 4. The flow-direction field therefore cannot be discarded as a drafting artefact. It identifies which geometric end corresponds to each flow-labelled invert.

The project must assemble the graph, use the flow-direction field to orient pipe ends, test shared-chamber continuity, and only then calculate grade and upstream/downstream reachability. A dataset with missing or unpopulated flow direction cannot be safely oriented from the invert values alone.

7.4 Field completeness in the Walkerville extract

Table 9: Usable values in all 1,002 Walkerville gravity-main records.

Field	Usable records	Interpretation
START_INVE	100%	Upstream invert under the utility flow convention.
END_INVERT	100%	Downstream invert under the utility flow convention.
GRADE	100%, one zero	Usable after orientation; range 0–9.61%.
NOMINALDIA	100%	150–450 mm; nominal diameter is not automatically internal hydraulic diameter.
FLOWDIRECT	100%	Coded 1 / 2 / 0; essential for interpreting which geometric end is upstream.
CONST_YEAR	99.5%	1896–2017; usable age feature with five missing values.
MATERIAL	55%	Mostly vitrified clay where known; 456 records unknown.
ROUGHNESS	0%	All values are zero; roughness must be assumed or calibrated separately.
Maintenance-hole surface level	10 of 360	Depth is effectively absent as a direct maintenance-hole attribute.

The Walkerville source is clipped to a council boundary. A pipe continuing into a neighbouring council can be cut mid-network, making an internal pipe appear to terminate and hiding downstream consequences. The statewide pipe layer removes this administrative clipping for its pipe extent, but its lack of maintenance holes means that a complete candidate-location graph still requires client or restricted access-point data.

7.5 Depth can be reconstructed with measured error

Maintenance-hole depth is cover level minus invert level. Because pipe inverts are available and contour lines supply a surface approximation, the updated assessment measured contour-interpolated cover levels against surveyed values.

Table 10: Contour-interpolated cover-level error reported in the updated data assessment.

Comparison	n	Mean abs.	P90	RMSE
1 m contours versus surveyed cover levels on the utility manhole layer	10	0.23 m	0.41 m	0.29 m
5 m contours versus surveyed levels, external validation set	707	1.34 m	2.62 m	2.08 m
5 m contour subset versus the full 1 m layer on the same terrain	1,009	1.85 m	4.54 m	2.59 m

The one-metre contour comparison lands within about a quarter of a metre on average, against chamber depths with a median of approximately 2.9 m. The external comparison used a South Australian council scheme that records surveyed surface, invert, and depth together only as a validation reference; it is not treated as a utility-network substitute. Records outside the contour footprint and records failing the publisher's own depth = surface minus invert identity were excluded before measuring. That identity held to 0.018 m across 726 passing cases. The practical conclusion is that fine contours can make depth reconstruction viable, while coarse five-metre contours degrade quickly and can approach half the quantity being estimated.

7.6 What still cannot be done from public data

The updated evidence leaves four material limits:

1. candidate sensor locations cannot be enumerated statewide because no public source lists maintenance-hole points for the catchment-complete pipe service;
2. roughness must be assumed because all examined public roughness fields are unusable;
3. placement cannot be validated against real chokes or overflows because no public source records historical events for this network; and
4. field-label conventions must be verified before graph construction because a wrong interpretation reverses upstream and downstream conclusions invisibly.

7.7 Restricted SA Waterservices and the updated data request plan

The following services were visible in published map definitions but requested authentication when their underlying data were checked. No credentials were used, offered, or sought.

Table 11: Access-restricted services and updated request priority.

Service	Layer(s)	Value and request
https://sawgis.sawater.com.au/arcgis/rest/services/AquaMap/NetworkReticulation/MapServer	49 maintenance holes; 42 gravity main; 46 connections; 52 inspection points	Layer 49 is the highest-priority request because it is the missing candidate-location set. Request surface level and depth wherever recorded for a defined study area.
https://maps.sawater.com.au/arcgis/rest/services/AssetWorksManagement/ServiceRequest/FeatureServer	3 wastewater service requests	Highest analytical value for validation. Request historical service requests, choke or overflow records with location and date.
https://sawgis.sawater.com.au/arcgis/rest/services/NetworkReticulation/Wastewater/MapServer	22 gravity mains; 35 structures	Useful for assurance, currency, and asset completeness; lower priority because statewide gravity mains are now openly downloadable.
https://services3.arcgis.com/ugMYGpcYkB7ZZz3l/arcgis/rest/services	Instrument layers visible in catalogue	Confirms that instrument layers exist but require login. The publicly listed instruments relate to drinking water, so request confirmation of whether a wastewater instrument layer exists and whether it contains locations only or time-series readings.

The request order is therefore: maintenance holes with levels; incident history; sensor-layer clarification and time series if available; and a client network extract for assurance of completeness and currency over the defined study area. Asking again for general network geometry alone is lower value because the open statewide service already provides a catchment-complete pipe layer. Endpoint addresses and layer numbers can change and live-service counts should be re-measured before they are used in a formal submission [Paul Jaison, 2026].

7.8 Data availability conclusions

The updated position is stronger than the earlier public-data assessment. A statewide gravity-main service now provides a public catchment-complete pipe extent with usable levels, while the Walkerville republication provides the most useful public maintenance-hole and contour context. Depth is no longer only a proposed reconstruction: fine-contour interpolation has a measured error. However, the two most important operational gaps remain: no public statewide candidate maintenance-hole layer and no public wastewater sensor or incident time series. The project can therefore build a public graph benchmark and a measured depth-reconstruction method, but client data are still required for candidate enumeration, real-event validation, current-sensor comparison, and operational recommendations.

To bridge this data gap during the initial research and algorithm development phase, open international smart-sewer benchmark datasets offer a rigorous testbed. Most notably, the Urban Water Observatory (UWO) dataset from Fehraltorf, Switzerland [Blumensaat et al., 2026] provides three years of continuous field laboratory observations across 124 sensors (including 89 wireless LoRaWAN level nodes) along with calibrated hydrodynamic models. Together with the 10-year continuous Bellinge sewer dataset, these open repositories allow the team to evaluate sensor placement algorithms, test sparse flow-reconstruction models, and verify resilience against missing telemetry before applying the finalized workflow to South Australian client data.

The method must also preserve the updated field semantics. It must use flow-labelled inverts and the flow-direction field together, validate shared chambers, report disconnected or boundary-limited components, assume roughness explicitly, and keep the statewide pipe graph separate from any inferred maintenance-hole candidate set.

8 Integrated implications for the SA Waterproject

8.1 Proposed purpose-specific workflow

The combined evidence supports the following workflow. It is a project method to be implemented and tested, not a completed client deployment.

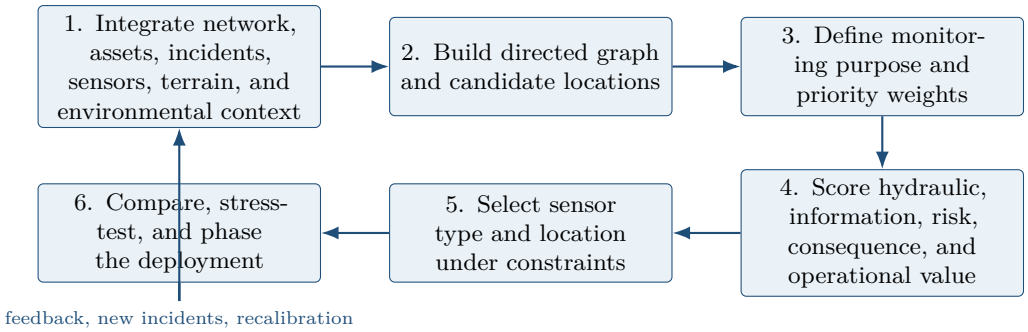


Figure 1: Proposed end-to-end placement workflow for the project.

8.2 Stage 1: integrate network data and environmental context

The first stage establishes the spatial and operational foundation from client network extracts. While public Walkerville data serve as a reproducible benchmark during method development, formal placement decisions require authenticated utility records. Integration tasks comprise:

- **Identity and topology preservation**: preserve persistent pipe, node, and chamber identifiers across all GIS feature classes;
- **Geometric snapping and validation**: snap coincident pipe vertices using a documented spatial tolerance (calibrated to SA Watercadastral precision) and audit all geometric repairs;
- **Flow-orientation interpretation**: interpret flow-labelled invert fields (`START_INVE`, `END_INVERT`) alongside the discrete flow-direction attribute (`FLOWDIRECT`), validate shared-chamber hydraulic continuity, and compute pipe slopes strictly after flow orientation;

- **Asset classification**: explicitly separate gravity mains, rising mains, pump stations, gravity outfalls, maintenance holes, and inspection openings;
- **State encoding**: preserve missing, zero, and unmeasured attribute values as distinct data states rather than imputing arbitrary defaults;
- **Operational record linkage**: join historical service requests, CCTV inspection grades, choke incident logs, and telemetry time series to specific pipe segments and manhole IDs across defined temporal windows; and
- **Hydrologic boundary definition**: delineate full catchment boundaries, flag disconnected or orphaned subnetworks, and record boundary-crossing mains that exchange discharge with adjacent jurisdictions.

Graph-reconstruction workflows demonstrated in DPP-SP provide a reliable template for addressing coincident endpoints, virtual junctions, duplicate chambers, and disconnected components, ensuring that hydraulic graph connectivity reflects true conveyance physics [Moradi and Najafi, 2026].

8.3 Stage 2: build directed graph and candidate locations

Once raw geospatial features are verified, Stage 2 constructs the directed hydraulic graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ and identifies admissible candidate sensor locations $\mathcal{C} \subseteq \mathcal{V}$.

8.3.1 Directed graph assembly

Edges $e = (u, v) \in \mathcal{E}$ represent sewer pipes oriented along the validated direction of gravity flow from upstream manhole u to downstream manhole v. Nodes $u, v \in \mathcal{V}$ represent manholes, junctions, pump stations, and outfalls. Adjacency and reachability matrices are generated to compute upstream tributary paths and downstream conveyance routes. While gravity networks are typically directed acyclic graphs (DAGs), cyclic structures introduced by siphons, bifurcations, or loops must be explicitly verified and handled.

8.3.2 Candidate location generation and physical screening

Candidate sensor locations begin with the universe of mapped access points: maintenance holes, pump sumps, drop shafts, and outfalls. Rather than treating every node as equally viable, physical and operational siting criteria filter out unsuitable structures prior to optimization:

- **Hydraulic stability (ASCE MOP 60 §3.7.2)**: Level and velocity instruments require stable, non-surcharged gravity flow during dry weather. Candidate manholes must provide straight pipe runs of 5 to 10 pipe diameters upstream and downstream, avoiding turbulent hydraulic jumps, sharp bends, steep drop junctions, and standing backwater pools [ASCE and WEF, 2007].
- **Chamber depth and headspace clearance**: Radar, ultrasonic, and acoustic sensors require adequate vertical headspace between the sensor face, the pipe crown, and the chamber invert. Shallow manholes ($< 1.2\,\text{m}$) lack sufficient beam clearance, while excessively deep chambers risk signal attenuation or ladder obstruction. Depths are determined from surveyed covers or 1 m contour interpolation [Paul Jaison, 2026].
- **Surface accessibility and traffic safety**: Monitoring sites must be safely serviceable by field crews without requiring extensive traffic lane closures on major arterial corridors or hazardous confined-space entry. Chambers located in accessible verges, footpaths, or quiet residential easements are prioritized over high-traffic roadway centers.
- **Telemetry and power feasibility**: Underground chambers beneath heavy cast-iron covers attenuate RF signals. Candidates must reside within verified cellular (NB-IoT / LTE-M) or LoRaWAN coverage zones, ensuring reliable antenna transmission to utility SCADA without requiring external elevated masts [Blumensaat et al., 2026].
- **Spatial deduplication and reachability**: Adjacent chambers within a documented spatial tolerance (e.g., $< 10\,\text{m}$) are consolidated, and terminal head nodes that cannot observe any upstream network are flagged.

8.4 Stage 3: define monitoring purpose and priority weights

Sensor placement is not a universal coverage problem; its objective must reflect the specific operational question defined by SA Water[PG S2-22 project team, 2026b].

8.4.1 Evaluation baselines

To provide defensible benchmarks, two deployment scenarios are established:

1. **Pristine network baseline (no existing sensors)**: identifies the mathematically optimal initial sensor configuration across the entire network from an unconstrained state; and
2. **Augmented operational baseline (existing sensors fixed)**: fixes all current SA Wateroperational sensors as hard constraints ($x_j = 1$), optimizing the placement of incremental additions or evaluating the opportunity cost of legacy locations.

8.4.2 Multi-criteria stakeholder weighting

When multiple operational goals compete for a limited sensor budget, candidate nodes are assigned transparent, multi-criteria priority weights w_i:

- **Failure probability**: pipe deterioration likelihood derived from CCTV defect histories, asset age, material, and predictive machine learning or Bayesian network models [Ma et al., 2025a, Malek Mohammadi et al., 2020];
- **Consequence and severity**: trunk pipe diameter, hydraulic conveyance capacity, potential spill volume, and repair difficulty;
- **Environmental and receptor sensitivity**: proximity to environmentally sensitive receiving waters (such as the River Torrens, Adelaide Parklands, and coastal outfalls) and public exposure corridors, following risk-weighted licensing frameworks [WSAA and Sydney Water, 2019]; and
- **Catchment hierarchy**: downstream reachability, upstream connected population, and tributary pipe count.

8.4.3 Operational criteria for early-warning purpose

An early-warning monitoring layout must be evaluated against operational criteria that go substantially beyond topological node coverage or geometric spread. As demonstrated in both the Stonyfell trial [Do et al., 2023] and Sydney Water's proactive blockage management [Drenoyanis and Prackwieser, 2022, Sydney Water, 2024], the utility value of an alert depends on whether it provides actionable lead time to dispatch maintenance crews before an overflow spills into customer properties or receiving waters. Consequently, the placement objective should score candidate locations based on:

- **event recall and missed-event rate**: the proportion of historical or simulated choke events successfully detected by the layout;
- **warning lead time**: the duration between the initial abnormal signal (such as an FID flag) and the hydraulic overflow, targeting the 8-to-57-day operational window observed in Stonyfell;
- **false-alarm and review burden**: the operational cost of false positives, ensuring alarm generation does not exceed utility triage capacity (avoiding the 100,000-alarm burden generated by uncalibrated static thresholds);
- **continuous data availability and telemetry uptime**: resilience against sensor dropouts, transmission latency, and battery exhaustion under diurnal logging constraints;
- **hydraulic position and reachability**: pairing upstream anomaly sensors with downstream confirmation points to establish double-threshold directional localization;
- **field accessibility and emergency response time**: ensuring prioritized monitoring locations can be safely accessed by jetting and clearing trucks without requiring extended traffic control or hazardous confined-space entry; and
- **receiving-environment sensitivity**: weighting candidates according to downstream receptor vulnerability, public exposure, and foot traffic, following the risk-weighted licensing principles established under NSW EPA PRP 307 [WSAA and Sydney Water, 2019].

This evaluation logic bridges the anomaly-detection method with the placement optimization: the algorithm must place continuous level monitors at chambers that maximize choke detection lead time while constraining anticipated false alarms across the operational network.

8.5 Stage 4: score hydraulic, information, risk, consequence, and operational value

In Stage 4, each admissible candidate chamber $i \in \mathcal{C}$ is evaluated across four core analytical dimensions prior to combinatorial optimization:

8.5.1 Hydraulic state and capacity scoring

Candidate locations differ substantially in their hydraulic sensitivity:

- **Elevation-head condition**: Following Ninh et al. [2025], sensor coverage capacity is position-dependent and governed by the physical elevation-head inequality:

$$Z_0 + H_d > Z_s + H_s,$$

  where Z_0 and Z_s are the invert elevations of the upstream monitored node and candidate sensor chamber, respectively, and H_d and H_s represent dry-weather flow depths. A downstream sensor can reliably monitor backwater surcharges only if the upstream invert remains hydraulically coupled; candidates on steep drops or with severe vertical breaks possess restricted upstream coverage horizons.

- **Conveyance headroom and Manning capacity**: Using topology-derived flow inference (TDFI) [Wang et al., 2026], full-pipe capacity $Q_{\text{Manning}} = \frac{1}{n} A R^{2/3} S_0^{1/2}$ is calculated across all incoming conduits. Chambers where peak wet-weather or I&I tributary inflows approach full conveyance capacity receive elevated monitoring priority.

8.5.2 Information-theoretic and state reconstruction value

Where flow or hydraulic state reconstruction is the primary monitoring objective, candidate nodes are scored on their ability to resolve unmonitored dynamics:

- **Expected Information Gain (EIG)**: Bayesian optimal experimental design (OED) calculates the mutual information between prospective sensor observations at node i and unmonitored state predictions across the network [Cheng et al., 2026].
- **Modal basis representation**: Following data-driven sparse sensing (DSS) [Ding et al., 2025], singular value decomposition (SVD) of spatiotemporal hydraulic matrices extracts the dominant empirical modes, and QR factorisation with column pivoting evaluates the numerical leverage of each candidate chamber.
- **Graph perceptual domain**: For graph neural network (GNN) architectures, candidate chambers are scored by their Ideal Perceptual Domain (IPD), bounding the theoretical receptive field across multi-hop graph convolutions [He et al., 2026].

8.5.3 Structural and failure risk scoring

Candidate chambers and adjacent conduits are evaluated for deterioration and blockage risk:

- **Bayesian network failure probability**: The learned multi-source BN structure evaluates the marginal defect probability $P(\text{Defect}_k \mid \text{Age}, \text{Dia}, \text{Mat}, \text{Soil})$ for each conduit [Ma et al., 2025a].
- **Historical choke clustering**: Spatial kernel density estimation maps historical choke clusters, recurring fatberg formations, and tree-root intrusion hotspots, providing an empirical likelihood score for localized sewer chokes [United States Environmental Protection Agency, 2005, Moradi and Najafi, 2026].

8.5.4 Operational feasibility and multi-criteria suitability

Finally, candidates are evaluated using a GIS-based fuzzy Analytical Hierarchy Process (AHP) [Nguyen et al., 2020]. Spatial criteria – including chamber invert depth, surface terrain slope, cellular signal strength, traffic volume, and distance to environmentally sensitive watercourses – are standardized through fuzzy membership functions, generating a composite operational suitability score $s_i \in [0, 1]$.

8.6 Stage 5: select sensor type and location under constraints

Stage 5 executes the combinatorial selection of sensor types and locations. Because different sensor technologies answer distinct operational questions (Table 12), the decision formulation assigns discrete sensor types $t \in \mathcal{T}$ (e.g., continuous radar level, float switch, paired acoustic, soft flow sensor, H_2S gas) to candidate chambers $i \in \mathcal{C}$.

8.6.1 Mathematical optimization formulation

Let $x_{i,t} \in \{0,1\}$ be a binary decision variable indicating whether sensor type t is deployed at candidate location i. The placement problem is formulated as a mixed-integer multi-objective optimization problem:

$$\max_{x} \quad [f_{\text{coverage}}(x),\ f_{\text{info}}(x),\ -f_{\text{search}}(x),\ -f_{\text{cost}}(x)] \tag{1}$$

$$\text{subject to} \quad \sum_{t\in\mathcal{T}} x_{i,t} \le 1 \quad \forall i \in \mathcal{C}, \tag{2}$$

$$\sum_{i\in\mathcal{C}} \sum_{t\in\mathcal{T}} c_{i,t}\, x_{i,t} \le B, \tag{3}$$

$$\text{dist}(u,v) \le D_{\max} \quad \forall (u,v) \in \mathcal{P}_{\text{monitored}}, \tag{4}$$

$$x_{j,t_{\text{existing}}} = 1 \quad \forall j \in \mathcal{S}_{\text{current}}, \tag{5}$$

where $c_{i,t}$ denotes the total life-cycle cost (capital acquisition, confined-space installation, telemetry subscription, battery replacements, and routine cleaning over a 5-year horizon), and B is the utility budgetary ceiling.

Constraint (2) ensures at most one instrument is installed per chamber unless complementary instruments (e.g., level and gas) are explicitly paired. Constraint (3) bounds total life-cycle expenditure. Constraint (4) enforces an operational spacing ceiling between consecutive sensors along any main conveyance line: as demonstrated by Li et al. [2023], inter-sensor spacing must not exceed $D_{\max} = 100\,\text{m}$ to guarantee double-threshold blockage localization. Constraint (5) preserves existing utility sensors. Additional constraints can mandate paired acoustic sensors in adjacent manholes for pipe headspace blockage pre-screening [Ninh et al., 2026].

For small-to-moderate candidate sets, the formulation is solved using mixed-integer linear programming (MILP). For large-scale directed utility graphs, the Evolutionary Greedy (EG) algorithm [Wang et al., 2024a] or NSGA-II [Wang et al., 2024b] is applied to generate Pareto-optimal trade-off frontiers with high computational efficiency.

Table 12: Purpose-specific sensor choices suggested by the evidence.

Operational purpose	**Candidate measurement**	**Placement and validation emphasis**
Early surcharge or overflow warning	Continuous radar or ultrasonic level; float/tilt for a simple threshold.	Rate of rise, trend persistence, alarm lead time, false alarms, condensation, fouling, and event capture.
Proactive blockage pre-screening	Paired acoustic sensors in adjacent maintenance holes [Ninh et al., 2026].	Headspace acoustic attenuation, frequency dispersion, dry-weather low-flow screening before backwater surcharge.
Network-wide hydraulic state	Level plus geometry, paired-manhole soft sensor [Lin et al., 2025b], or model-reconstructed flow.	Hydraulic model calibration, NSE/NMSE/RMSE, peak timing, volume, backwater, and node-level as well as system-level error.
Local choke or abnormality	Continuous level trend and anomaly detector (FID).	Normal-pattern training, seasonal and wet-weather separation, 14-day rolling baseline, detection delay, and inspection confirmation.
Inflow and infiltration (I&I) tracking	Topology-derived flow inference (TDFI), level plus Manning capacity, pump SCADA.	Subcatchment attribution, full-pipe capacity exceedance (Q_{Manning}), diurnal baseflow separation, and rain response.
Contaminant source or spread	Water-quality or contaminant sensor plus hydraulic transport model.	Source observability, reliability, decay, dilution, threshold, upstream/downstream trade-off, and false positives.
Odour, corrosion, or septicity	Headspace gas or dissolved H_2S according to the control question.	Phase, concentration range, humidity, ventilation, dosing point, calibration, and maintenance.
Structural or operational condition	CCTV, acoustic, computer vision, or fibre inspection.	Defect labels, inspection frequency, access, image quality, and agreement with maintenance outcomes.

8.7 Stage 6: compare, stress-test, and phase the deployment

The final stage benchmarks the optimized layouts against reference baselines, subjects selected designs to operational stress-testing, and schedules deployment across phased investment tranches.

8.7.1 Comparative benchmarking against baselines

To isolate algorithmic gains from underlying network topology, the optimized layout is compared against five standardized baselines under identical budget and candidate constraints:

1. **Current utility placement**: existing SA Watersensor layout, establishing actual operational performance;
2. **Seeded random placement**: 1,000 Monte Carlo random draws reporting mean performance and 95% confidence bounds;
3. **Standard engineering heuristics**: rule-based placement targeting hydraulic outfalls, major trunk confluences, pumping station wet wells, and environmentally sensitive discharge points;
4. **Topological centrality screening**: ranking candidates by graph betweenness and in-relevance harmonic centrality [Simone et al., 2023]; and
5. **Greedy information / risk coverage**: greedy selection maximizing cumulative mutual information [Crowley et al., 2025] or weighted risk set cover [Moradi and Najafi, 2026].

Table 13: Evaluation measures for the proposed project experiments.

Measure	Question	Use
Coverage or weighted risk coverage	How much of the selected objective is observed?	Upstream nodes, high-risk pipes, environmental weights, or population; report denominator and weights.
Mutual information and redundancy	How much unique information does the set contain?	Compare repeated versus complementary sensors; report time window, preprocessing, and noise assumptions.
Reliability and detection rate	How often is a simulated event detected?	Contaminant, overflow, or threshold scenarios; report event set and threshold.
RMSE, NMSE, R^2, NSE	How well are unmonitored states reconstructed?	Flow or level prediction; report system-level and node-level performance.
Peak and response error	Is the warning timely and operationally useful?	Peak magnitude, time-to-peak, rate-of-rise, detection delay, and missed events.
Search and response cost	How hard is it to locate or investigate an alert?	Downstream/upstream search effort and expected manual inspection.
Resilience	What happens when a sensor is noisy, missing, or failed?	Single and multiple dropout, telemetry loss, battery failure, and re-optimised versus fixed designs.
Life-cycle cost	Is the layout maintainable?	CapEx, installation, battery, telemetry, cleaning, calibration, access, and replacement.

8.7.2 Robustness stress-testing

Real-world sewer monitoring environments are hostile and imperfect. Candidate layouts must undergo rigorous sensitivity and failure testing:

- **Sensor dropout sensitivity**: evaluating performance degradation when individual or clustered sensors fail. As observed in Ding et al. [2025], system-level averages can conceal extreme vulnerability: while most dropouts reduce NSE by < 0.05, the failure of a single critical trunk sensor can degrade reconstruction accuracy by over 36%;
- **Measurement noise and fouling**: injecting 5%, 10%, and 15% multiplicative Gaussian noise and drift to simulate transducer fouling, acoustic reflections, and foam interference;
- **Telemetry latency and packet loss**: evaluating early-warning lead times under constrained telemetry cadences (e.g., 6-hour or 24-hour batch uploads versus real-time alarms), verifying whether battery-preserving logging preserves sufficient lead time for emergency jet-truck dispatch; and
- **Hydraulic parameter uncertainty**: perturbing pipe Manning roughness coefficients and simulating extreme storm events (e.g., 10-year to 100-year ARIs) to ensure placement stability under shifting hydraulic regimes.

8.7.3 Phased deployment scheduling and adaptive feedback

Following Ninh et al. [2025] and Ma et al. [2025b], utility rollouts should proceed in phased capital investment tranches rather than a single static deployment:

- **Phase 1 (Critical trunk and choke screening)**: Deploy continuous level sensors at primary hydraulic outfalls, major pumping interfaces, and historical choke hotspots, establishing baseline diurnal flow signatures and confirming telemetry reliability.
- **Phase 2 (Secondary network densification)**: Install paired acoustic pre-screening sensors [Ninh et al., 2026] and intermediate level monitors along high-risk branches, bounding inter-sensor spacing to $\leq 100\,\text{m}$.
- **Phase 3 (State reconstruction and model integration)**: Deploy soft flow sensors [Lin et al., 2025b] and GNN reconstruction models [He et al., 2026] to estimate water levels across unmonitored lateral connections.

Crucially, as shown in the outer feedback loop of Figure 1, operational observations from Phase 1 feed back into the workflow: newly recorded choke incidents, verified dry-weather diurnal baselines, and observed telemetry packet-loss rates update the Bayesian network priors and fine-tune subsequent placement stages.

9 Challenges, assumptions, and limitations

9.1 Data access and completeness

The central unresolved dependency is access to a catchment-complete SA Waternetwork extract, incident history, and historical sensor readings. Without these, the project can still produce a reproducible public-data benchmark and method prototype, but it cannot claim client validation. The public survey also shows that topology alone is easier to obtain than the attributes needed for hydraulic, condition, and life-cycle decisions.

9.2 Hydraulic and model limitations

Several placement methods use simulated data. A hydraulic model can provide a complete state matrix when sensors are sparse, but its output inherits assumptions about roughness, inflow, rainfall, storage, pump operation, and boundary conditions. A topology-only method avoids some model costs but misses hydraulic behaviour. The correct response is not to select one method universally; it is to compare model-dependent and model-light methods and report their disagreement.

9.3 Sensor and deployment limitations

Sensor placement is constrained by conditions underground. Condensation and foam can affect ultrasonic or optical devices. Solids and grease can obstruct floats. Flow estimates need geometry and calibration. Gas measurements depend on phase and ventilation. Telemetry and power can turn a mathematically good location into an operationally poor location. Safe access and maintenance frequency may remove high-scoring candidates.

9.4 Statistical and algorithmic limitations

The reviewed studies use different definitions of coverage, reliability, error, risk, and information. Neural networks can fit simulated patterns without transferring to field data. Bayesian networks depend on local factor relationships and condition labels. Greedy methods are fast but can miss a better combination; genetic and tabu methods explore more but need reproducibility controls. Mutual information can favour clusters when all nodes are equally weighted. Sparse sensing can produce excellent system-level error while leaving individual nodes less accurate.

9.5 Boundary, safety, and governance

The analytical boundary must be a hydraulic catchment rather than an arbitrary council boundary. Historical data may include customer, location, or operational information. Access, safety, cultural constraints, and physical installation approvals must be obtained from the responsible authority; they should not be inferred from map data. Code, data transformations, and results should be peer reviewed before a recommendation is handed over. The project is a research precursor to later implementation and is not an operational approval or a field-validated deployment decision [PG S2-22 project team, 2026b,a].

Table 14: Known challenge, effect on the placement study, and response.

Challenge	Effect	Required response
Network extract delayed or incomplete	No client-validated topology, attributes, or candidate set.	Run a public benchmark; request catchment-complete assets; replace benchmark data when approved.
Missing depth and surface	Weak surcharge, grade, and installation interpretation.	Use fine contours where available; report the measured interpolation error; request maintenance-hole levels where the contour interval is too coarse.
Missing roughness or internal diameter	Flow and capacity calculations become assumption-driven.	Keep nominal and internal fields separate; run material-based and sensitivity scenarios.
No incidents or historical sensors	No direct test of detection, lead time, or current placement.	Request dated events and time series; use synthetic scenarios only as clearly labelled validation.
Boundary clipping and disconnected components	Upstream coverage and outfall interpretation may be wrong.	Rebuild catchments, report components, and include boundary-crossing assets.
Sensor noise or dropout	A high-average layout may fail when a critical sensor is unavailable.	Test noise and dropout; add redundancy or fallback sensors where needed.
Life-cycle cost unknown	CapEx-only ranking may be operationally misleading.	Include installation, telemetry, battery, cleaning, calibration, access, and replacement.
Cross-network transfer	Published percentages may not hold in South Australia.	Treat external results as method evidence; re-run all outputs on SA Waterdata.

10 Current research progress and next work

10.1 Research completed in the current record

The team has established:

- a wastewater-specific problem definition and scope boundary;
- a sensor technology map covering level, flow, water quality, H_2S, condition, and telemetry;
- a comparative placement-method review with information, topology, optimisation, ML, risk, and sparse-sensing methods;
- a condition and failure-factor catalogue;
- an open-data inventory for South Australian utility and statewide wastewater networks;
- a detailed list of restricted SA Waterservices and the fields needed from them; and
- a proposed end-to-end workflow, baseline comparison, metric set, and robustness plan.

10.2 Information still needed from the client or project environment

The next stage requires decisions and data rather than more generic literature:

1. the target catchment or pilot area and its analytical boundary;
2. the current sensor inventory, sensor types, readings, sampling intervals, and known outages;
3. the preferred monitoring purpose and evaluation priority;
4. a catchment-complete network extract with asset identifiers, node relationships, pipe geometry, and elevations;
5. dated incident, choke, overflow, maintenance, CCTV, and complaint records with locations where available;
6. hydraulic model files, calibration information, or representative level and flow series;
7. installation, access, telemetry, power, safety, and maintenance constraints; and
8. verified CapEx and OpEx assumptions for each sensor class.

10.3 Proposed deliverables from the compiled evidence

The research record supports the project deliverables already identified in the scoping material: literature review; data and algorithm report; processed dataset; reproducible code; evaluation results; recommendations; and a documented handover. The final recommendation should include the data version, preprocessing, candidate rules, weights, budget assumptions, method settings, baselines, metrics, uncertainty tests, and limitations so that a different analyst can reproduce the result.

11 Conclusion

The team evidence supports a clear direction for the wastewater sensor-placement project. A useful monitoring network is not defined by the largest number of sensors or by a single universal algorithm. It is defined by the operational question, the network processes that connect locations, the consequences of missed events, and the ability to maintain and use each measurement.

The sensor review shows why level, flow, water quality, gas, condition, and telemetry technologies must be selected for their specific purpose. The placement literature shows that information, redundancy, topology, hydraulic reliability, source-identification value, risk, search effort, cost, and reconstruction accuracy are different objectives. The condition literature provides a practical feature catalogue, with age, diameter, length, material, soil, traffic, environmental, and operational evidence all potentially relevant. The South Australian data survey shows that a public topology benchmark is possible, but client data are required for complete catchment analysis, incident validation, sensor time series, and operational cost.

The appropriate project output is therefore a transparent, purpose-specific, mixed-sensor optimisation workflow. It should compare current, random, rule-based, topology, greedy, and multi-objective layouts; use the best available client evidence; preserve missingness and uncertainty; test noise and sensor failure; and report coverage, information, detection, reconstruction, response, resilience, and life-cycle cost together. The final sensor density, locations, and expected improvement should be produced by that evaluation on the selected SA Watercatchment.

A Glossary of network and monitoring terms

Table 15: Working glossary used in the compiled paper.

Term	Meaning in this project
Gravity main	A sewer pipe that conveys wastewater under gravity and is normally part-full.
Rising main	A pressurised sewer pipe associated with pumping.
Maintenance hole	An access point and common candidate location for a sensor.
Invert	The elevation of the inside bottom of a pipe at an endpoint or structure.
Surface level	Ground or cover elevation near an asset.
Depth	Surface level minus invert; it should not be confused with wastewater level.
Grade	Change in invert elevation along a pipe after the flow-labelled endpoint convention has been interpreted.
Roughness	Hydraulic resistance parameter, such as Manning's n; a field with no usable values cannot support a calibrated roughness calculation.
Node and edge	A graph location and directed connection.
Upstream set	Nodes whose conveyed flow can reach a selected sensor under the network direction.
Downstream set	Nodes or outlets reachable after a selected node.
DAG	Directed acyclic graph; acyclicity must be demonstrated from an assembled, correctly oriented graph rather than assumed from the invert-field inequality.
Catchment	The hydraulically connected area draining to a defined outlet or treatment path.
Coverage	A defined measure of how much of the selected objective is observed; it is not automatically geographic area.
Reliability	In the contaminant studies, the fraction of simulated events or injection points detected under a specified threshold.
Observability	The ability to infer unmeasured states or identify a source from sensor observations.
NSE	Nash-Sutcliffe efficiency for reconstruction agreement; values closer to 1 indicate closer agreement under the chosen test.

Term	Meaning in this project
I&I	Infiltration and inflow entering a sewer system from groundwater, stormwater, or other unintended paths.
CWMS	Community wastewater management system.

B Detailed study matrix

Table 16: Methods, data, findings, and transfer limits from the principal reviewed studies.

Study	Data and network	Method and metric	Finding and transfer limit
Zheng et al. 2025	878-node urban drainage network; operating state and flood classification.	VOI and TE for location; NMF for matrix-completion evaluation; error and classification accuracy.	Four sensors: 33% error and 76% accuracy; eight: 29% and 82%. River section required separate treatment; costs were illustrative.
Li et al. 2012	Brazos, 12 gauges; Delfland, 1,520 candidates.	Joint entropy, transinformation, total correlation; MIMR; duplicated information.	Top six Brazos stations under 20% duplicate information versus 32–72% for WMP; Delfland top nine about 85% information. Needs dense time series and no direct cost.
Crowley et al. 2025	479-node south England network; about 1,020-node Bellinge network; two-minute flow/depth data.	Mutual information; modified greedy; Gaussian assumption; GLM and GRNN; NMSE.	Selected layouts generally beat random and simple rules; gains depend on sensor count and predictor; clustered layouts and equal node weights need correction.
Cheng et al. 2026	2,576 candidate locations in Detroit corridor; tRIBS-Urban flood simulations.	Bayesian optimal experimental design (OED); expected information gain (EIG); DNN surrogate.	Parameter-oriented design does not equal prediction-oriented design; spatial rankings shift with lead time and target. Feasibility constraints must bound pure EIG.
Simone et al. 2023	77 nodes/79 links and 531 nodes/530 pipes; SWMM contaminant scenarios.	SCIP global optimisation; binary detection; in-relevance harmonic centrality; reliability.	Four sensors reached full reliability at one threshold in the 77-node case; 4, 8, and 11 in the 531-node case at different thresholds. Topology is a screen, not a hydraulic substitute.
Wang et al. 2024a	4,394-node Hong Kong network and synthetic benchmarks.	Evolutionary Greedy (EG) algorithm; cost-coverage-resolution; runtime and hypervolume.	Matches NSGA-II solution quality while scaling substantially faster for large directed networks. Highly valuable as a rapid screening solver.

Study	Data and network	Method and metric	Finding and transfer limit
Wang et al. 2024b	Tuen Mun, 4,394 nodes and 4,308 edges; more than 500,000 residents.	NSGA-II; cost-coverage-resolution; hypervolume, AMA, STMA.	About 20 sensors gave 80% coverage in the trade-off set; optimised layouts beat human heuristics. Flow, access, and field validation were not included.
Salem and Abokifa 2024	84 conduits, 80 junctions, 59 sheds; 150 two-source events.	PySWMM; MLP concentration forecasting; GA placement; observability and reliability.	Upstream improves reliability, downstream observability; adding a third sensor improved reliability 16% and observability 1.8%. Simulation and decay assumptions limit transfer.
Ninh et al. 2025	Two real gravity sanitary sewer systems; multi-phase investment.	Genetic algorithm with position-dependent elevation-head condition ($Z_0 + H_d > Z_s + H_s$).	Direct link between physical invert, dry-weather depth, and coverage. Explicitly handles multi-phase deployment; coverage capacity varies by manhole position.
Ninh et al. 2026	Experimental pipe rig (DN150/DN225); acoustic wave transmission.	Paired acoustic sensors in adjacent maintenance holes; frequency attenuation and phase dispersion.	Proactively detects 10% to 60% cross-sectional blockage in pipe headspace before liquid surcharge occurs. Complements hydraulic level monitoring.
Huang et al. 2025	Two real urban drainage pipe networks; multiple rainfall events.	Graph partitioning + Bayesian experimental design + ensemble smoother model calibration.	Jointly places sensors and calibrates model parameters; outperforms flood hotspot and uniform layouts in parameter uncertainty reduction.
Wang et al. 2026	Real urban drainage network with heterogeneous sub-catchment flows.	Topology-derived flow inference (TDFI); SBS algorithm; pipe Manning capacity cap.	Individual pipe detection thresholds (IPDT) capture heterogeneous I&I flows; prioritising dense building areas significantly improves detection.
Nguyen et al. 2020	Ålesund, Norway; stormwa-ter/wastewater network.	GIS-based fuzzy Analytical Hierarchy Process (AHP); 8 spatial criteria.	Generates flood-sensor suitability map; manholes, altitude, and rainfall most sensitive. Candidate screening layer rather than algorithmic optimizer.
Mohammadi et al. 2020	23 sewer condition studies (2001–2019) across international utilities.	Systematic review of physical, environmental, and operational sewer deterioration factors.	Confirms age, material, diameter, and length as core physical predictors; diameter effects are contradictory across catchments. Establishes feature engineering basis.
Ma et al. 2025a	23,000 CCTV-inspected sewer pipes in Hong Kong with GIS, traffic, soil, and climate.	Bayesian network (15 nodes, 31 links); mutual information, joint probability, true risk influence.	Pipe age (MI 0.124), diameter (0.032), population (0.027), and soil dominate; critical state probability up to 72%. Factor relationships provide prior risk weights.

Study	Data and network	Method and metric	Finding and transfer limit
Moradi and Najafi 2026	33,474 reconstructed graph elements; 48 existing stations; CCTV, maintenance, SSO labels.	RF/XGB/MLP risk; greedy, GA, tabu weighted set cover; ROC-AUC, PR-AUC, risk coverage.	RF plus greedy raised coverage from 11.73% to 32.26% at 48 sensors in the external case; radius and candidate ceiling dominated.
Moradi et al. 2025	Smart wastewater and stormwater collection systems overview.	Algorithmic taxonomy: exact, greedy, metaheuristic, evolutionary algorithms and AI/ML models.	Maps models to monitoring questions; exact solvers do not scale to city networks; underscores necessity of domain validation rather than generic AI labelling.
Ma et al. 2025b	Screened 189 studies (96 retained) across global wastewater and stormwater networks.	Comprehensive review of IoT sensor placement methods, objectives, algorithms, and evaluation metrics.	Categorizes operational vs abnormal monitoring; highlights hybrid optimization and phased deployment with feedback as superior to static single-objective runs.
Ding et al. 2025	Duluth Woodland, 77 nodes, 250 SWMM training scenarios; stormwater.	SVD reduced basis; QR pivoting; NSE; exhaustive and Monte Carlo benchmarks; noise/dropout.	Three sensors mean observed-event NSE 0.949; noise was modest but individual dropouts could be severe. Stormwater and unvalidated model.
He et al. 2025a,b	527 unmonitored drainage nodes; real rain events.	Spatiotemporal GCN (STGCN) and physical-attribute edge-enhanced GCN; prediction accuracy.	7 sensors gave 71.3% accuracy; adding invert elevations lifted predictions to 88.6% at 10% density. Physical edge features vastly outperform plain MLPs.
He et al. 2026	Large urban drainage network; demand-driven placement.	Ideal perceptual domain (IPD) metric vs actual perceptual domain (APD) with GNN reconstruction.	Targeting 90% IPD needed sensors at only 3.0% of nodes and achieved 86.7% APD coverage in real rain. Placement ignores pipe diameter in selection step.
Qi et al. 2025	C-Town benchmark drainage network; unlabelled spatial data.	GCN topological feature extraction + structured deep clustering network.	Over 94% agreement with expert labels; reduced monitoring points by >40% while covering >90% critical nodes without labelled training data.
Chen and Yu 2025	LID stormwater collection network; short-horizon dynamics.	Cluster-then-forecast framework; centroid selection + LSTM forward forecasting.	Monitored centroids predict water levels 5 min ahead across unmonitored network from 25 min history; shifts with nonstationary storm events.

Study	Data and network	Method and metric	Finding and transfer limit
Do et al. 2023	Stonyfell, South Australia; 11.3 km network, 62 operational sensor series, 1 year.	Ultrasonic level sensing; Finding Irregular Duration (FID); 14-day rolling window.	Warned before all 24 confirmed chokes with 32.6 days average lead time; static 150 mm threshold generated >100,000 alarms. Anomaly detector, not placement algorithm.
Drenoyanis and Prackwieser 2022	Sydney Water reticulation network; thousands of IoT devices; digital water meter trial.	Risk-based ten-step site selection; diurnal pattern comparison; paired chamber flow error.	20 blockages detected per month and about $400,000 per month avoided cost; daily upload prevents real-time paired alarms. Industry operational report.
Sydney Water WWOM 2024	Metropolitan Sydney network; 8,000+ level sensors (15k planned).	Active network sensing; I&I source control vs storage containment; PRP 307 receptor scores.	300–370 blockages/month caught before spills; storage containment (>A$18B, 350+ yrs) rejected for I&I source control (17x cheaper); risk-weighted licensing.
WSAA Case Studies 2019	Icon Water (CS7), Unitywater (CS6), Melbourne Water (CS1), Watercare (CS8).	Multi-criteria risk assessment; data-led SCADA screening; 1-in-5 yr ARI containment.	Icon Water saved A$173M using smart monitoring over pipe upsizing; Unitywater abated 90 hotspots via SCADA screening; 50% I&I from private laterals.
US EPA CMOM 2005	US municipal collection systems national assessment.	Federal Capacity, Management, Operation, and Maintenance (CMOM) compliance evaluation.	Blockages cause 48% of overflow events, but capacity limitations cause 75% of total overflow volume. Divergent reticulation vs trunk monitoring objectives.
ASCE MOP 60 2007	Gravity sewer design manual; instrumentation siting checklist.	Hydraulic criteria: straight run (5–10 D), undisturbed gravity flow, no backwater.	Non-contact level and flow instruments require non-surcharged gravity flow and safe access without confined space entry. Sets physical candidate filters.
Blumensaat et al. 2026	Fehraltorf, Switzerland (Urban Water Observatory); 124 in-sewer sensors (89 LoRaWAN).	3-year continuous catchment observation; water levels, temperature, runoff, calibrated SWMM.	High-density open benchmark for validating placement algorithms, sparse sensing, and telemetry resilience before field deployment.
Li et al. 2023	Rural DN150–DN300 sewers; pilot rig plus six full-scale level sensors; 0.1–0.2 kg blockages.	Double-threshold rule on level change rate and range; identification and positioning matrix.	95.3% identification within 4 m upstream, 80.4% and 95.6% overall; spacing should not exceed 100 m. Empirical thresholds, clean water in pilot.

Study	Data and network	Method and metric	Finding and transfer limit
Lin et al. 2025a	Three real manholes in China at 1 min for 2021, plus one CFD manhole.	MLP soft sensor on a symmetric level window; MRE and NSE; noise and bias tests.	MRE 2.88% and NSE 0.9372 baseline; robust to noise but amplifies flow-data bias. Needs a flow campaign per manhole.
Lin et al. 2025b	Two real pipes (600 mm/70 m; 1,200 mm/113 m) and one simulated pipe; Raspberry Pi 5 with LoRa.	Saint-Venant with HLL solver, MLP, and PINN on paired manhole depths; R^2, MRE; data-efficiency sweep.	SVE 13.0% MRE on one day of data with no wet weather; MLP more accurate with 35 days; PINN failed. Both pipes already had flow meters at both ends.
Wang et al. 2024c	Two Ningbo catchments, combined and separate; 1 min flow, 0.2 mm rain gauge.	Random forest and LSTM flow prediction; I&I as wet minus modelled dry weather flow; resolution sweep.	I&I volume error under 7%; accuracy falls from 5 to 60 min resolution. I&I is derived not measured; catchment-specific.
Zhang et al. 2020	Espoo, Finland; two pumping-station subcatchments; C-band radar rainfall.	ANFIS and MLPNN; three rainfall thresholds; RMSE and R^2; minimum dry weather flow screen.	Both subcatchments showed I&I; ANFIS better overall; minimum dry weather flow above zero is a model-free infiltration screen. Data not yet sufficient for full I&I quantification.

C Proposed project experiment grid

Table 17: Suggested experiment grid once the client dataset is available.

Experiment	Inputs	Comparisons	Primary outputs
Topology benchmark	Public or client graph, inverts, candidate nodes.	Current, random, outfall, confluence, centrality, greedy coverage.	Upstream coverage, component reach, candidate distribution, runtime.
Level event detection	Level series, rainfall or inflow context, incident labels.	Float threshold, continuous level anomaly (FID), current placement, optimised placement.	Event recall, false-positive rate, warning lead time, missed-event rate, alarm burden, data coverage, and verification delay.
Flow reconstruction	Calibrated or benchmark hydraulic model, geometry, level/flow series.	Random, information greedy, QR/DSS, GNN/STGCN, model-observability placement.	System and node NSE/NMSE/RMSE, peak and volume error.
Inflow & infiltration (I&I)	Rainfall series or radar, dry-weather diurnal patterns, pump SCADA, Manning capacity (Q_{Manning}).	Uniform detection threshold vs topology-derived flow inference (TDFI); SCADA screening vs paired-manhole soft sensors.	I&I detection accuracy, lead time to capacity exhaustion, false alarms, subcatchment attribution.
Risk placement	CCTV, condition, incidents, asset attributes, environmental receptor scores.	Existing, random, rule-based, RF/BN risk, weighted set cover.	ROC/PR/F1, weighted risk coverage, isolated risk, sensitivity to radius.
Mixed-sensor optimisation	Sensor type, cost, access, power, telemetry, maintenance.	Single-type versus mixed-type (level, paired acoustic, soft sensor); CapEx-only versus life-cycle objective.	Pareto front or weighted ranking, cost, coverage, redundancy, maintainability.
Resilience	Noise model, dropout patterns, telemetry gaps, missing fields.	Fixed layout, re-optimised layout, redundancy constraints.	Performance loss distribution, critical sensors, recovery strategy.

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